



ODONT 541.1

CARIOLOGY

Enamel Caries

Lecture #2

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(Tuesday and Wednesday afternoons)

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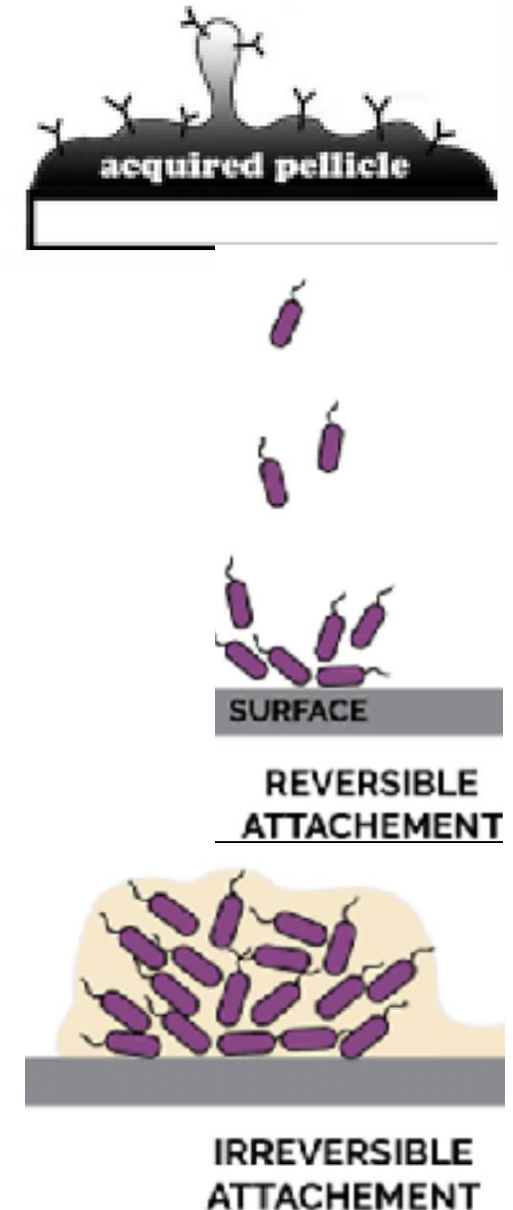
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Typo from last week

1. Acquired pellicle formation
 - Within seconds after a tooth is cleaned, salivary proteins, glycoproteins, lipids, phospholipids and carbohydrates adsorb to the enamel surface. This forms a thin, bacteria-free conditioning film called the acquired pellicle.
2. Initial Adhesion
 - Reversible attachment
 - Planktonic bacteria in the saliva loosely adhere to the acquired pellicle via weak forces. Predominantly Streptococci.
 - Permanent/Irreversible attachment
 - Bacteria utilize appendages like fimbriae and pilli and surface adhesins to create strong bonds to the pellicle.
 - Bacteria produce glucans/EPS excretion, which promotes stronger adhesion
 - Microbial succession
 - Initially, the biofilm bacteria were mostly aerobic organisms like streptococci. Over time, the plaque microbiota become more diverse.
 - As biofilm becomes thicker, oxygen concentration decreases. With microbial succession, more **anaerobes** like Acintomyces grow into the mix.
 - Over time, the plaque becomes mostly anaerobic bacteria. Carbohydrate fermentation is predominantly performed by anaerobic organisms.



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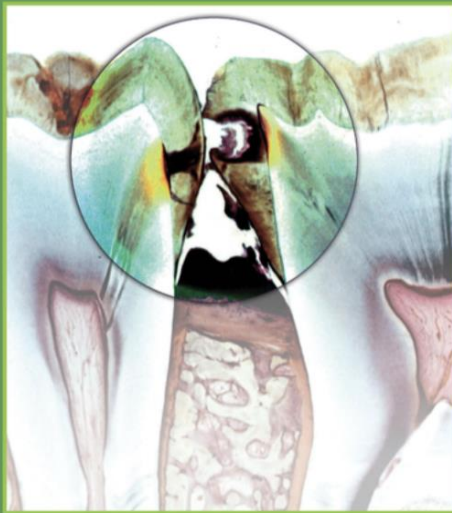
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Recommended Reading

Fourth Edition

Dental Caries

The Disease and Its Clinical Management



Edited by **Ole Fejerskov** and **Bente Nyvad**

WILEY Blackwell

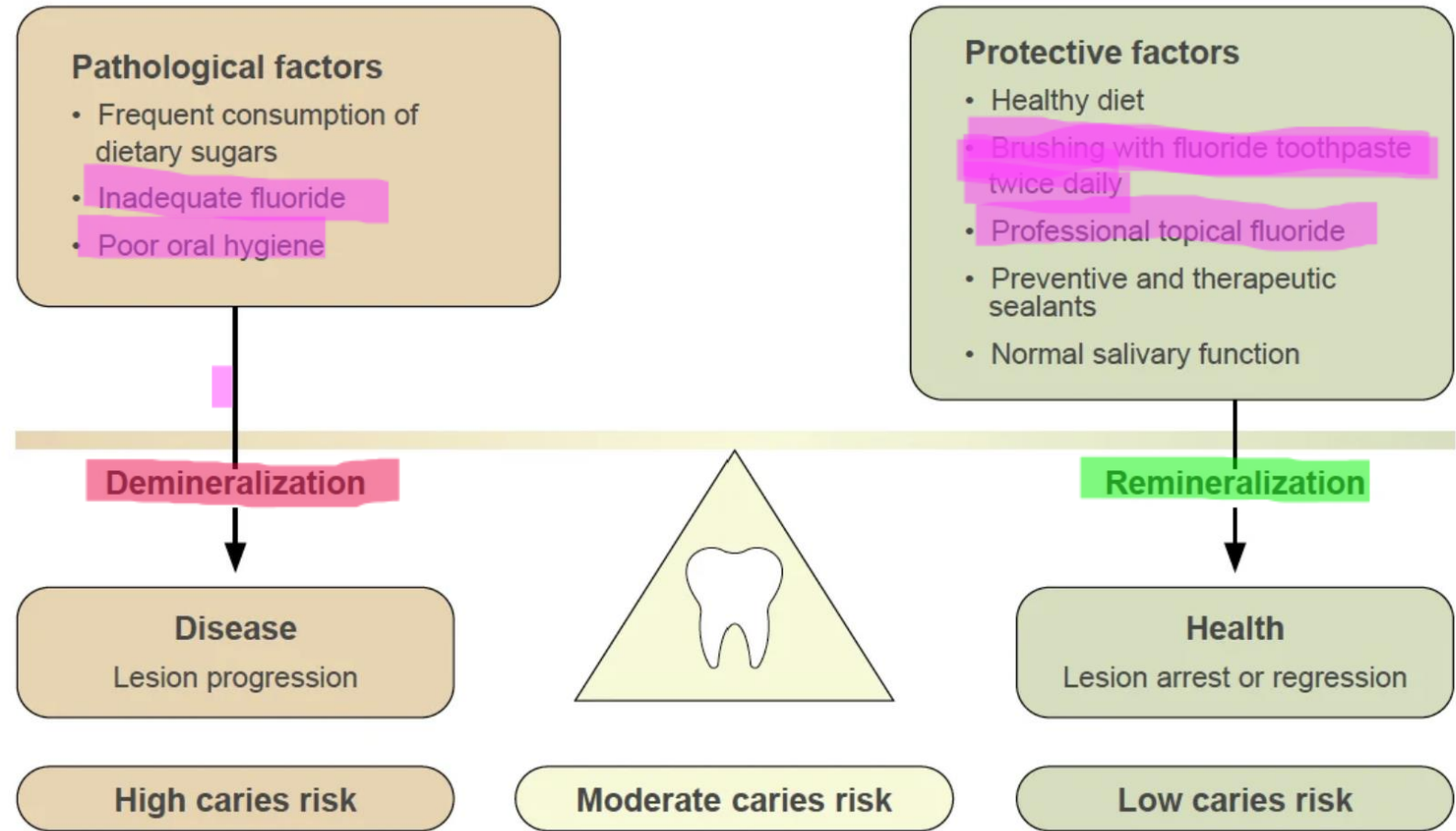
Textbook:
- Page 181-216

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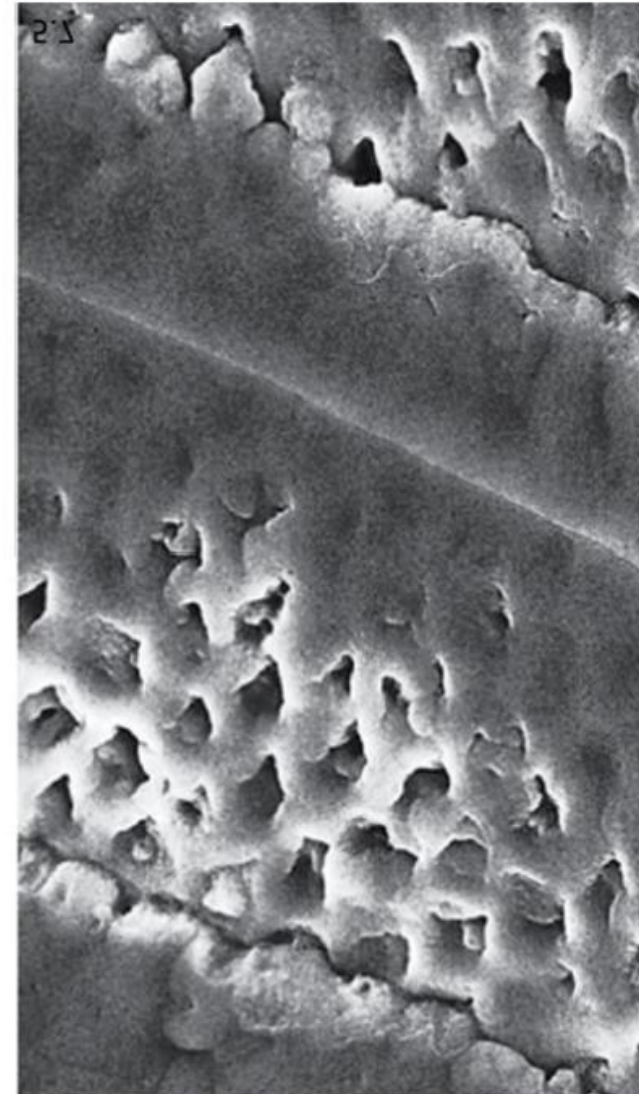
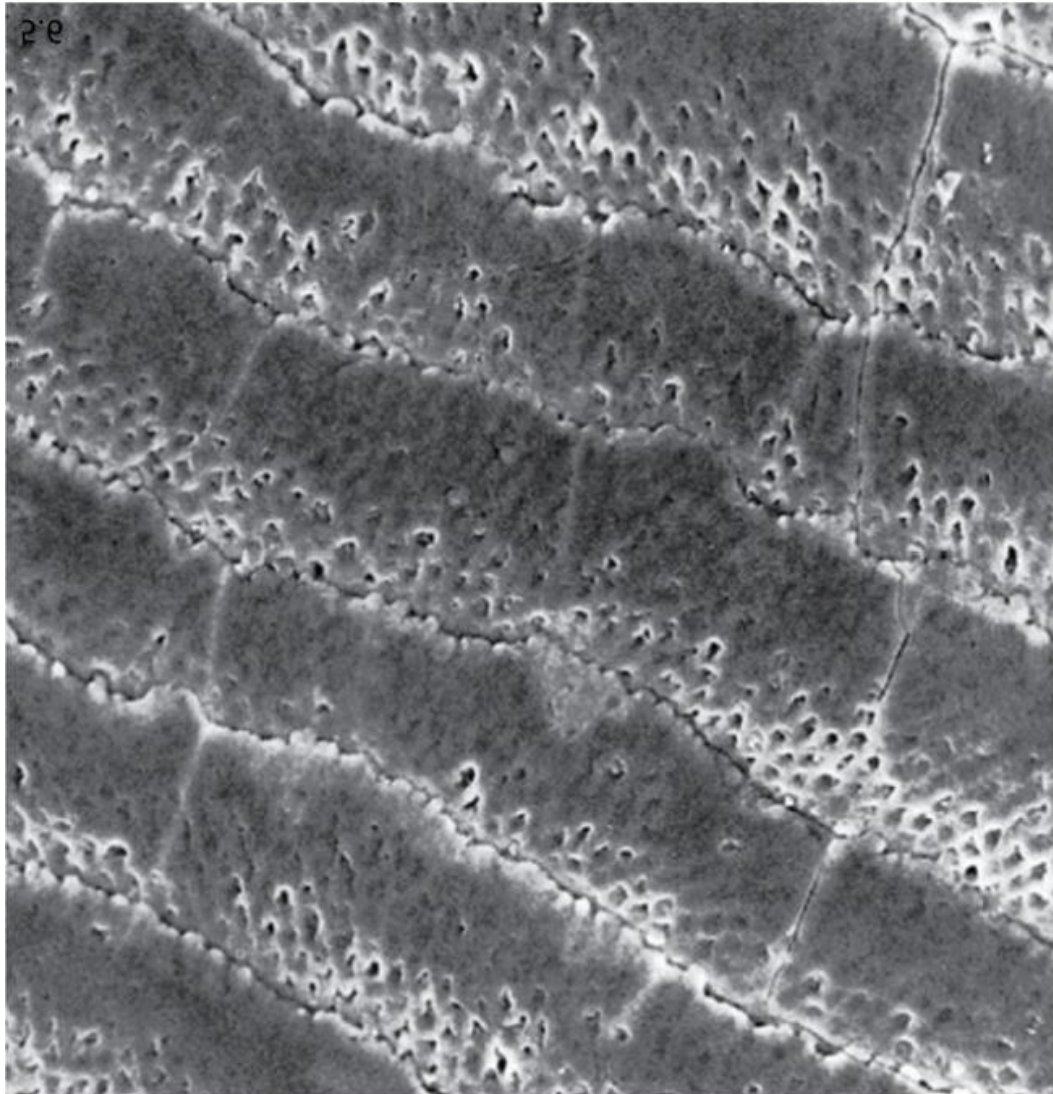
Pathologic and protective factors are responsible for the activity and rate of the biochemical reactions that eventually cause caries.

To understand caries, we must first understand the demin/remin biochemical reaction.

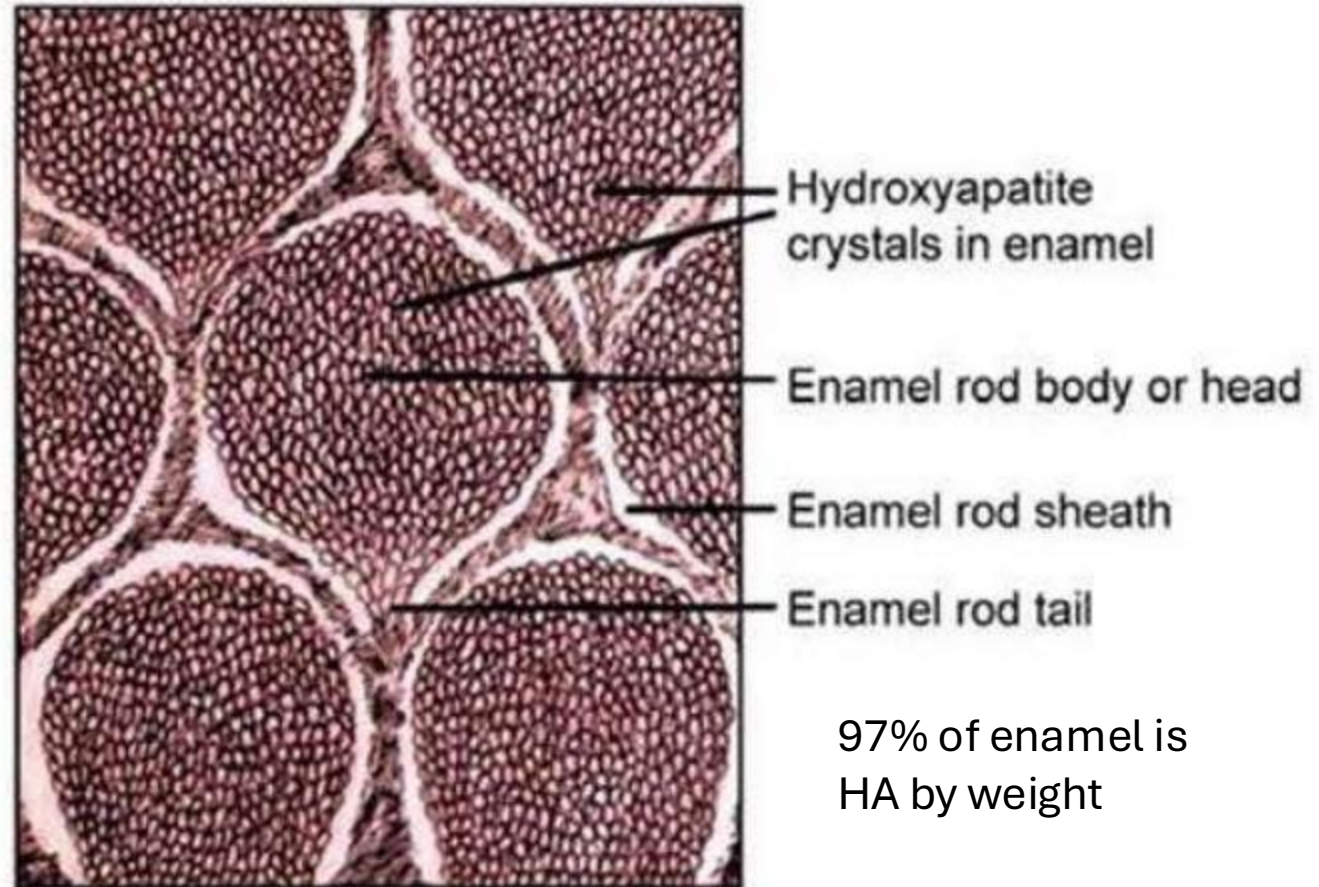
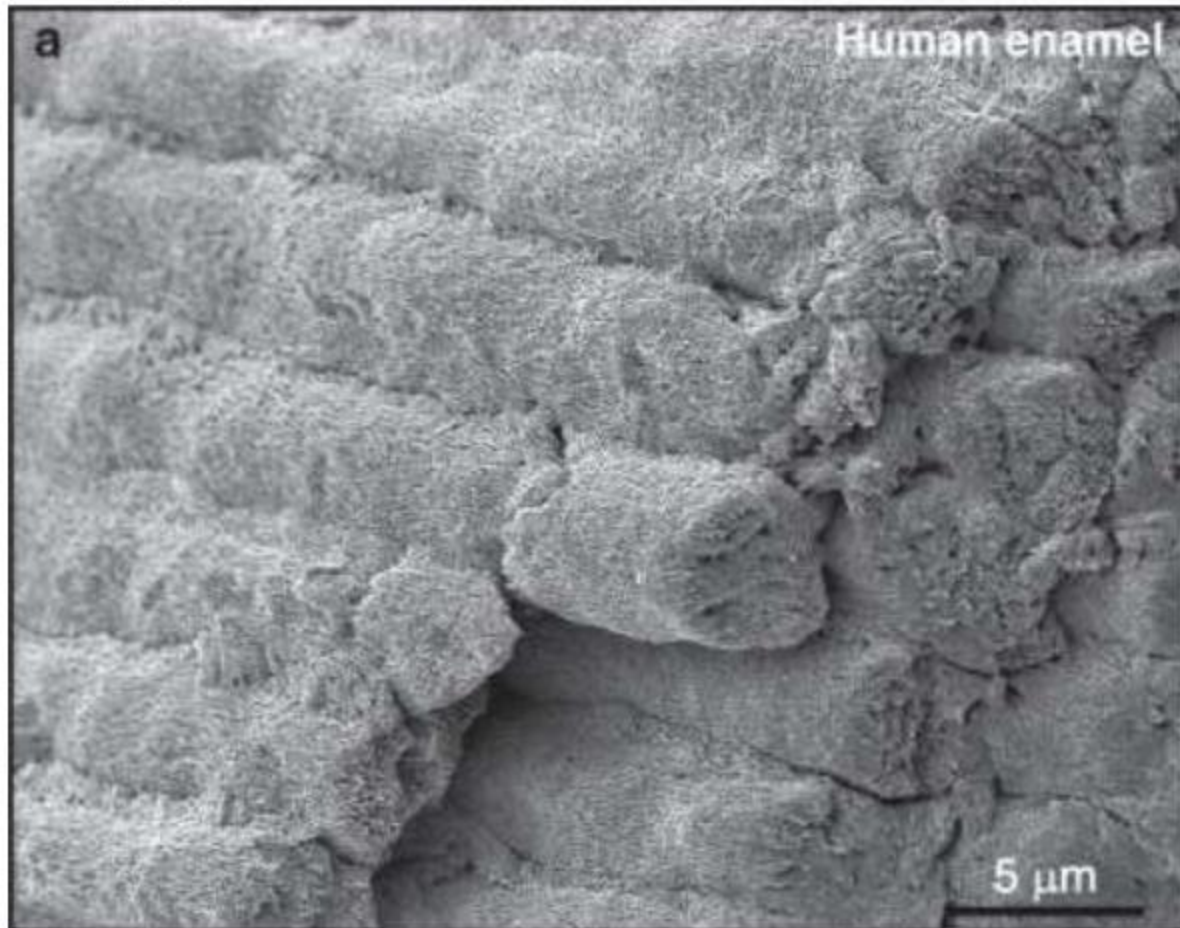
Then, we must consider how that reaction changes in different environments (how the pathologic and protective factors prevent or alter the reaction).



Caries starts on the outermost surface of the tooth,
which in a healthy tooth, should be enamel.
The outermost layer of enamel is very porous.



Enamel Structure



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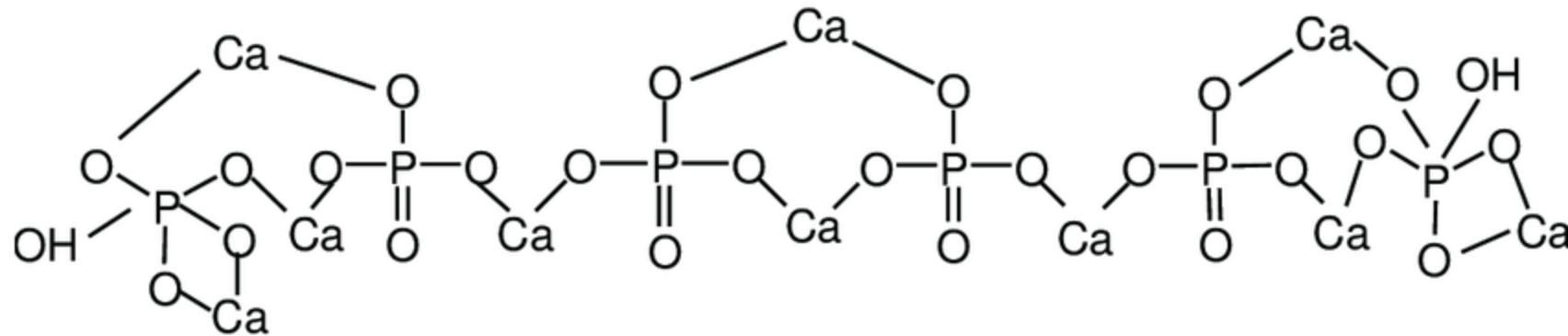
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Hydroxyapatite

HA is the predominant mineral in enamel. Multiple molecules of HA combine to form hexagonal crystals.

HA crystals, of varying size, are naturally found in:

- Teeth
- Bones
- Scales
- Shells (eggs, shellfish)
- Pathologic calcific conditions
 - Tendinitis, kidney stones, etc.
- Limestone



HA can also be synthesized for medical and industrial applications.

De/Re Mineralization

Fermentable carbohydrates + oral microflora → ACID (H⁺)



Hydroxyapatite (HA)

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Hydrogen Ions

pH is a logarithmic measure of free hydrogen ion concentration in an aqueous environment.

pH > 7 = Basic
pH < 7 = Acidic

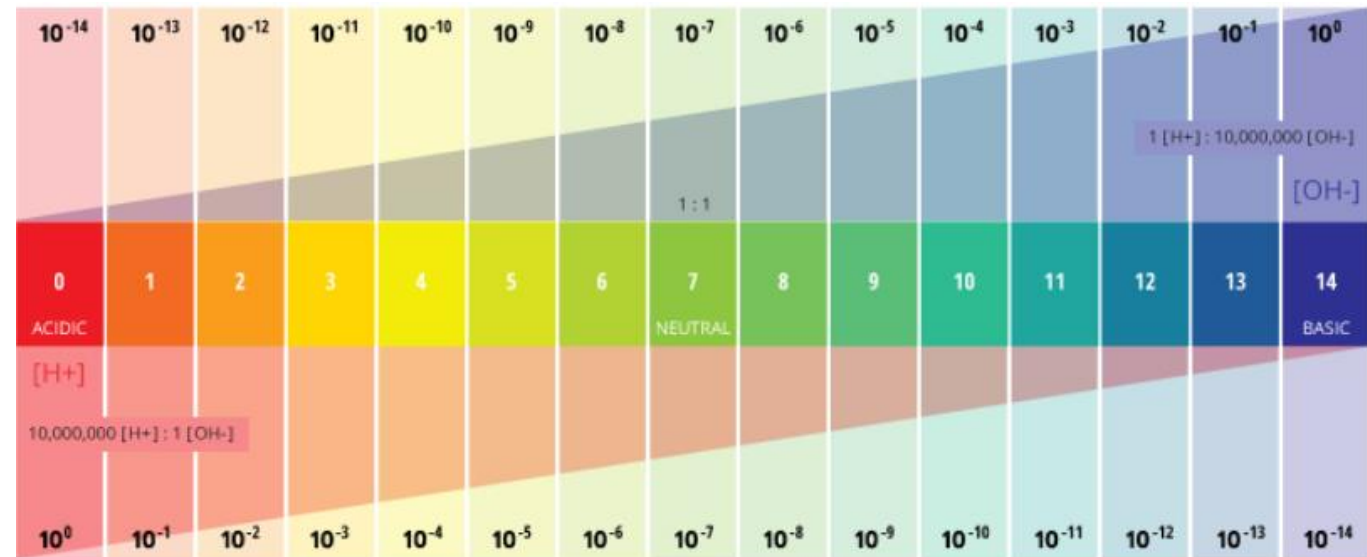
If the pH changes from 7 to 6, there are 10x more free hydrogen ions in the environment.

If the pH changes from 7 to 8, there are 10x more free hydroxide (OH-) ions in the environment.

Think of these ions as potential energy, like a battery that can drive a chemical reaction forwards.

The demineralization reaction advances when sufficient hydrogen ions are present. What type of oral environment has more free hydrogen ions present and would therefore promote more demineralization?

How would a change in pH from 5 to 4 alter the rate of demineralization?



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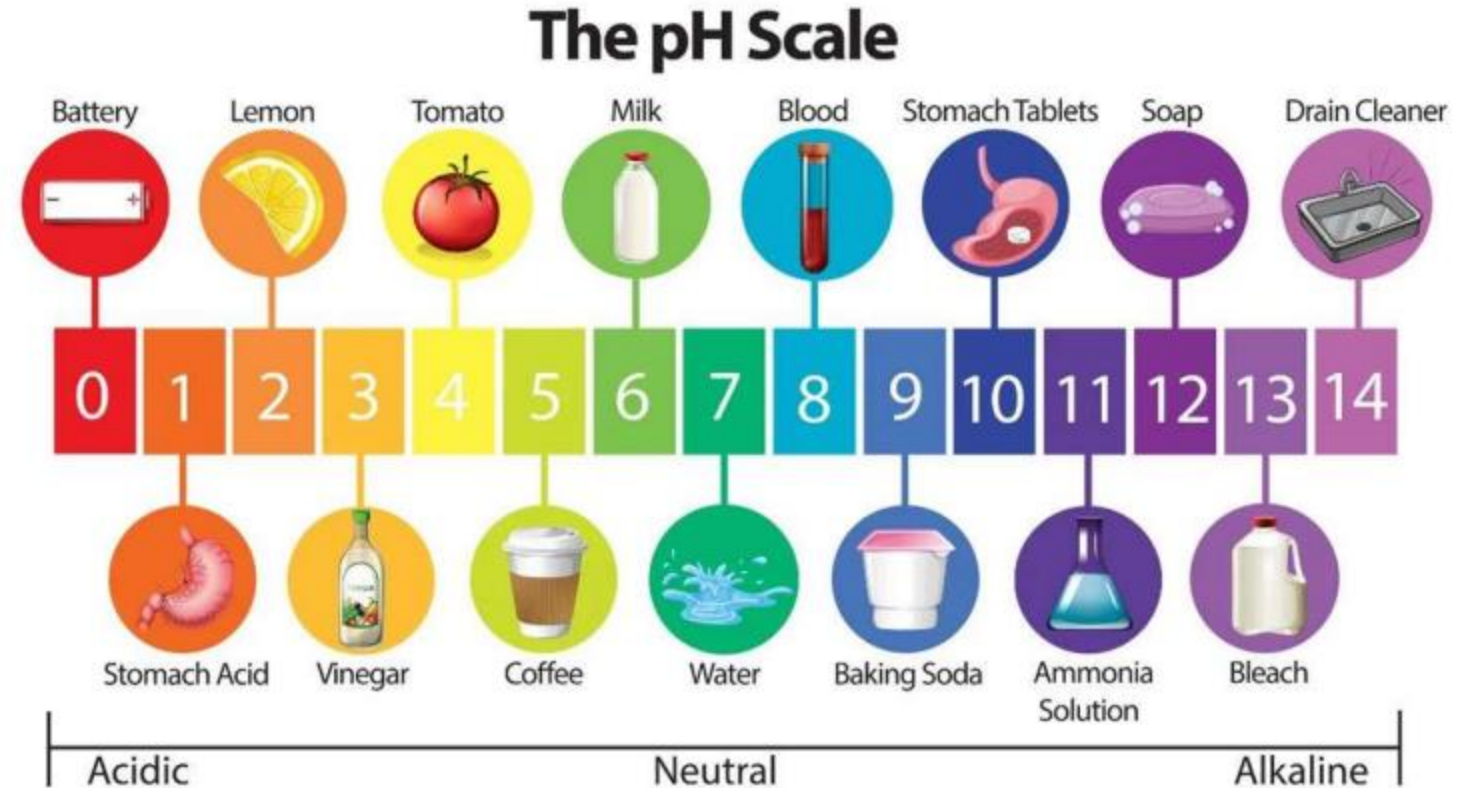
Hydrogen Ions

At physiologic pH, there is some biochemical activity occurring...but not too much.

Some activity is important for physiologic turnover.

Why do we need physiologic turnover in our bones and teeth?

If physiological turnover is a natural process and has some benefits, should we as dentist's seek to completely halt the demineralization reaction?



Physiologic oral pH 6.5-7.5 → IDEAL

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Fluorohydroxyapatite

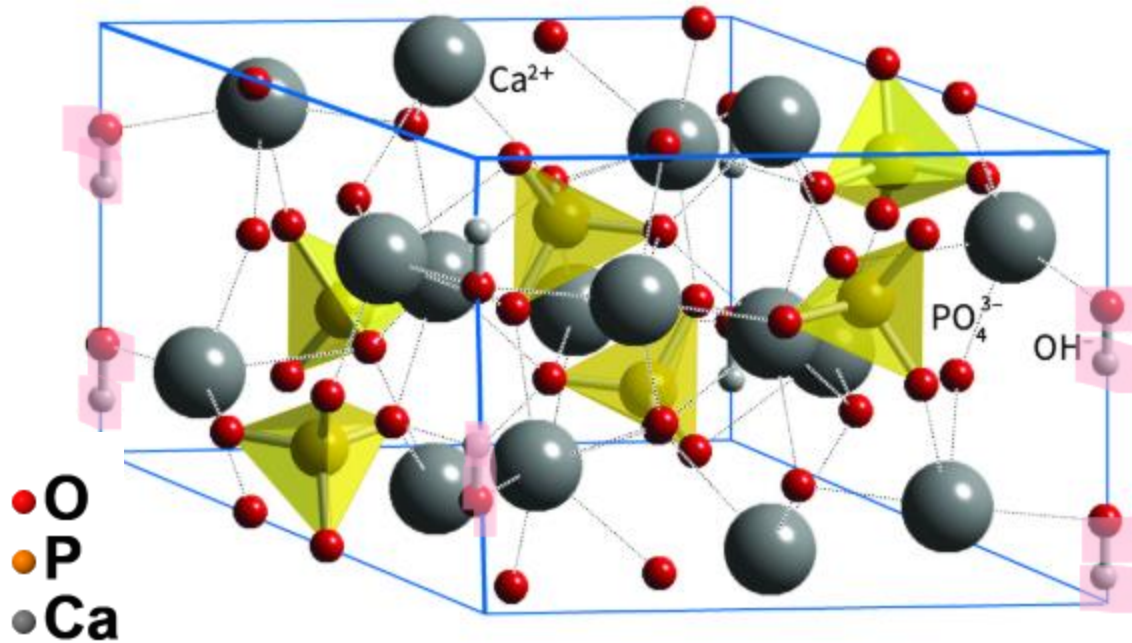


- Aka FA or fluoroapatite
 - $\text{Ca}_5(\text{PO}_4)_3\text{F}$
- Also naturally occurring, as natural background exposure to fluoride is common.
- Fluoride ion is more electronegative than hydroxide ion. Therefore:
 - F- ion is more compact (smaller) than OH- ion
 - FA is more compact/dense than HA
 - FA has higher crystallinity than HA
 - FA is more stable than HA (less likely to demin)
 - If fluoride and hydroxide are both present, the fluoride is more likely to bind. This will form FA instead of HA.
- FA will form more rapidly than HA would

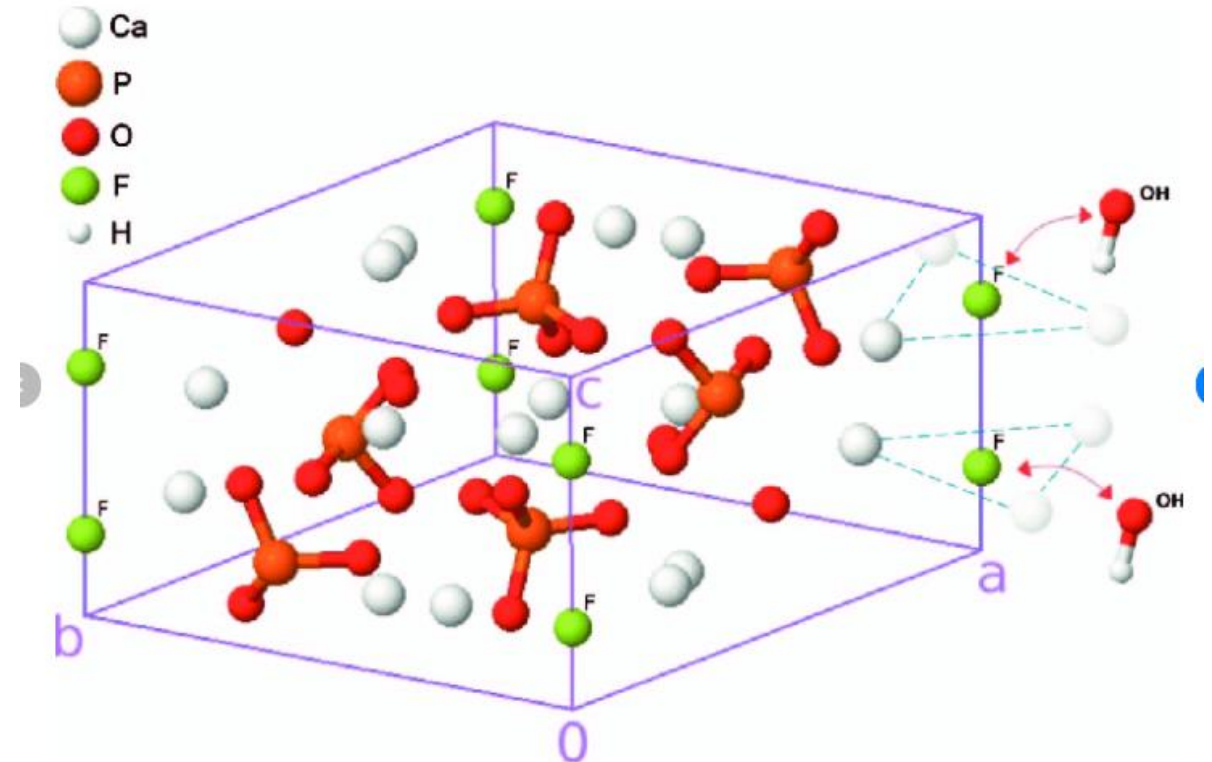
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HA vs FA



The structure of hydroxyapatite, $\text{Ca}_5(\text{OH})(\text{PO}_4)_3$, shown as Ca^{2+} ions coordinated by phosphate and OH^- ions into a strong three-dimensional structure. The isolated O atoms are OH^- with the H lying outside the unit cell.



Crystal unit cell of FA space group $P6_3/m$; $a = b = 9.368 \text{ \AA}$, $c = 6.884 \text{ \AA}$, $\alpha = \beta = \gamma = 90^\circ$, and $\alpha = \beta = \gamma = 120^\circ$. HA has the same hexagonal structure, differing only in the substitution of all fluoride ions by hydroxide ions.

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Healthy Oral Environment

- pH is 6.5-7.5
 - Demin/remin reaction is happening but at a slow rate and potentially balanced in both directions as the pH fluctuates between 6.5-7.5
- Oral fluid is **supersaturated** with both HA and FA
 - What does supersaturated mean?
 - The aqueous solution (saliva) is so full of minerals that any additional minerals will not dissolve but will instead precipitate out of solution and deposit onto available surfaces, sometimes in crystalline form.
- Calculus formation begins with the deposition of calcium and phosphate from supersaturated saliva.



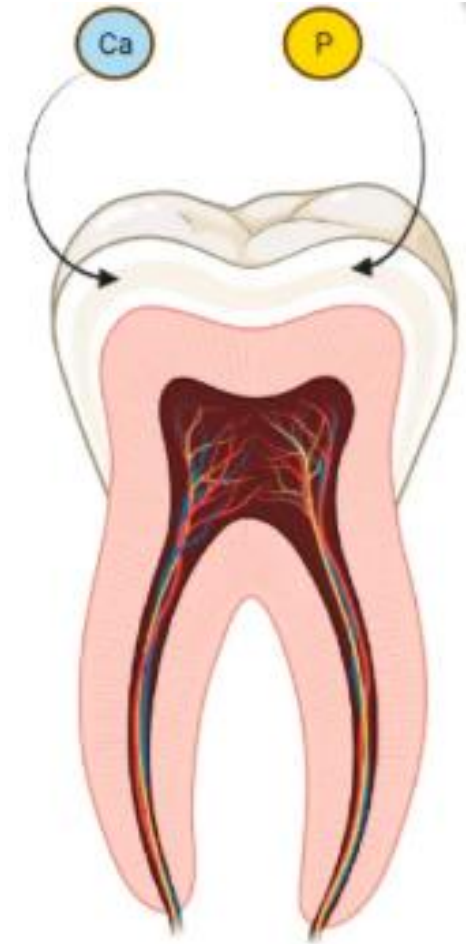
Demineralization

- “Unhealthy” oral environment starts with a significant change in pH
 - What caused that significant change in pH?
- The pH at which a substrate begins to demineralize is called it’s critical pH
 - The critical pH of normal, healthy, HA based enamel is **5.5**
 - The critical pH of FA based enamel is **4.5**
- At pH of 5.5, HA in enamel begins to dissolve but any FA is still stable and remains. FA does not begin to demineralize until the pH falls below 4.5.
- The dissolution of HA is the beginning of caries in enamel



Remineralization

- If the pH is between 4.5-5.5 AND the components of FA are present, remineralization occurs and FA is formed. The HA that demineralized is quickly remineralized to FA, as much as available building blocks allow.
 - What are the components of FA?
 - Even while HA is demineralizing, FA can remineralize. Because they have different critical pHs.
- If the pH is over 5.5, HA can remineralize. But FA will still preferentially form if F⁻ is present.
- If the pH is under 5.5 and no fluoride is present, the demineralized HA can not remineralize into FA
- If the pH is under 4.5, even FA will demineralize. No remineralization will occur.
- Remineralization requires BOTH the constituent parts of HA/FA and the appropriate environmental pH to drive the reaction



Remineralization

- What does a patient need in order to remineralize their teeth? And where do they get it?
 - Calcium
 - Where?
 - Phosphate
 - Where?
 - Anion (Hydroxide or Fluoride)
 - Where?
 - Which would you prefer for remineralization? How do they differ?
 - Amenable environment
 - More alkaline than critical pH
 - Building blocks can reach the site in need of remineralization
 - Time
 - Which type of remin occurs faster? HA or FA?
- How long does remineralized enamel last?
 - Only until the next time the critical pH is reached.

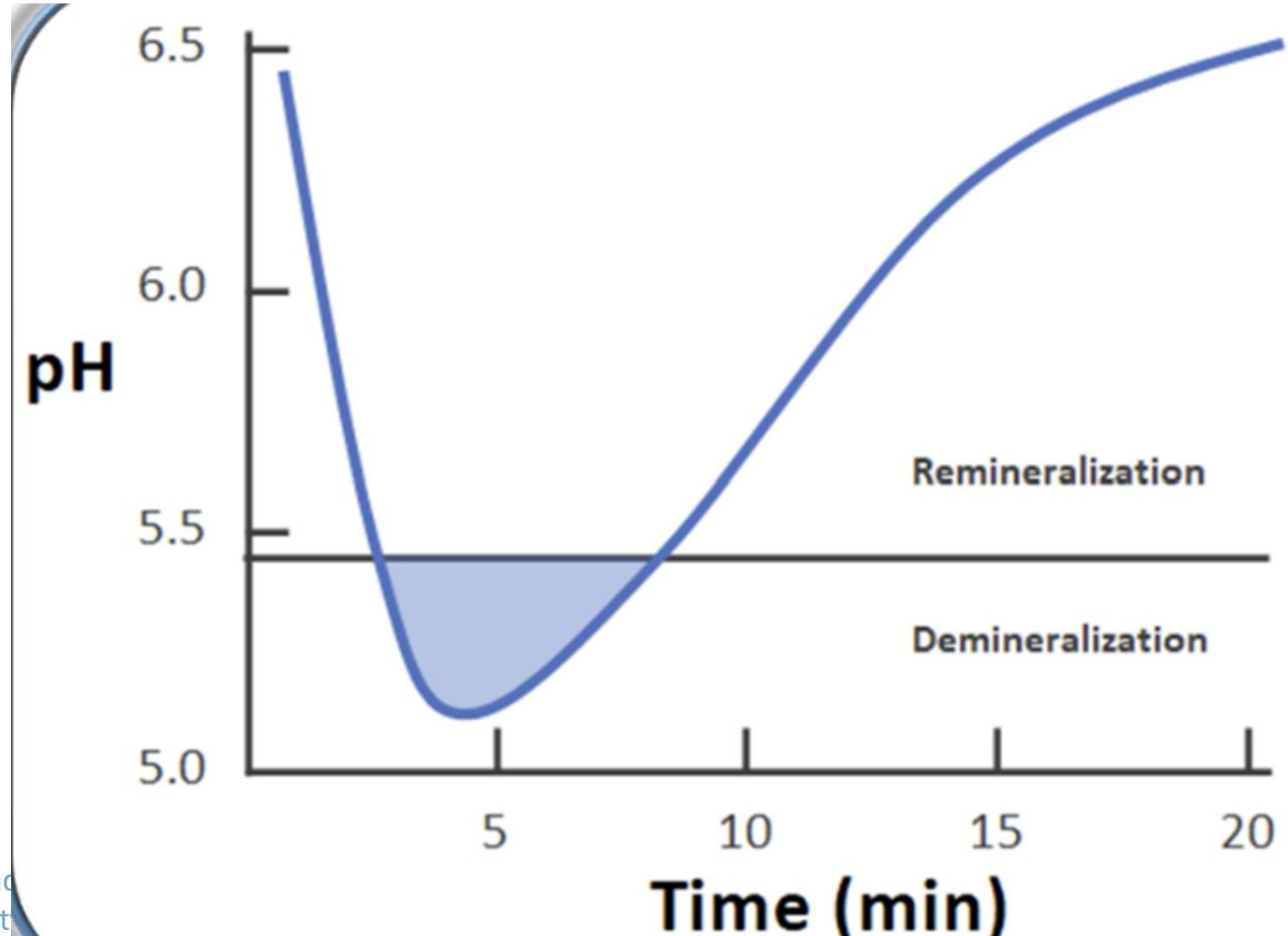
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Simplified representation of the change in pH, and shift in remin/demin balance, after a sucrose exposure.

Does this curve drop below 4.5?

Stephan's Curve



Salivary pH vs Biofilm pH

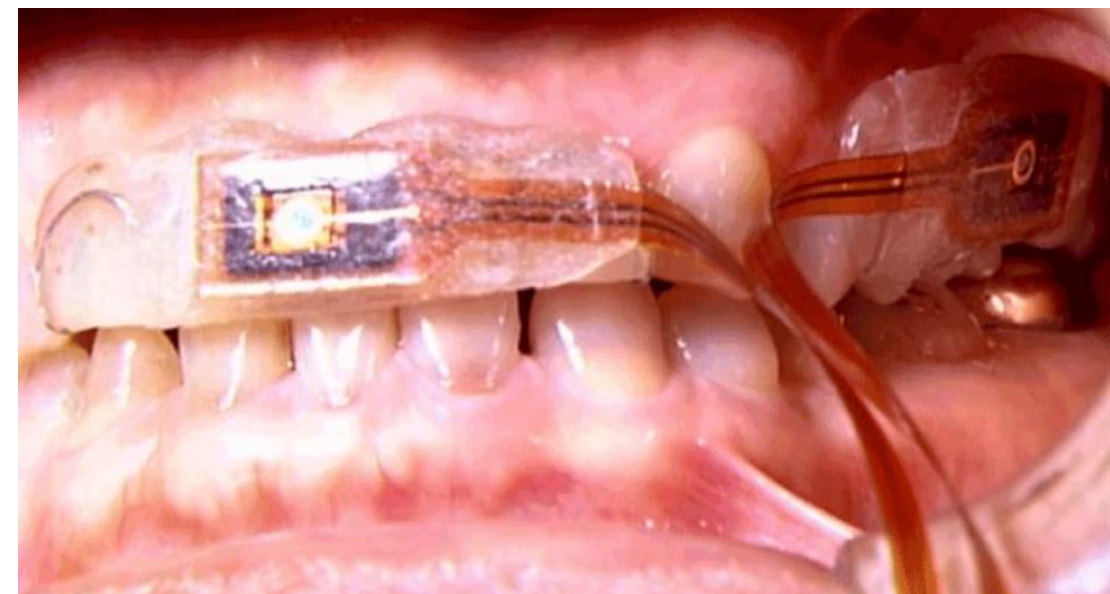
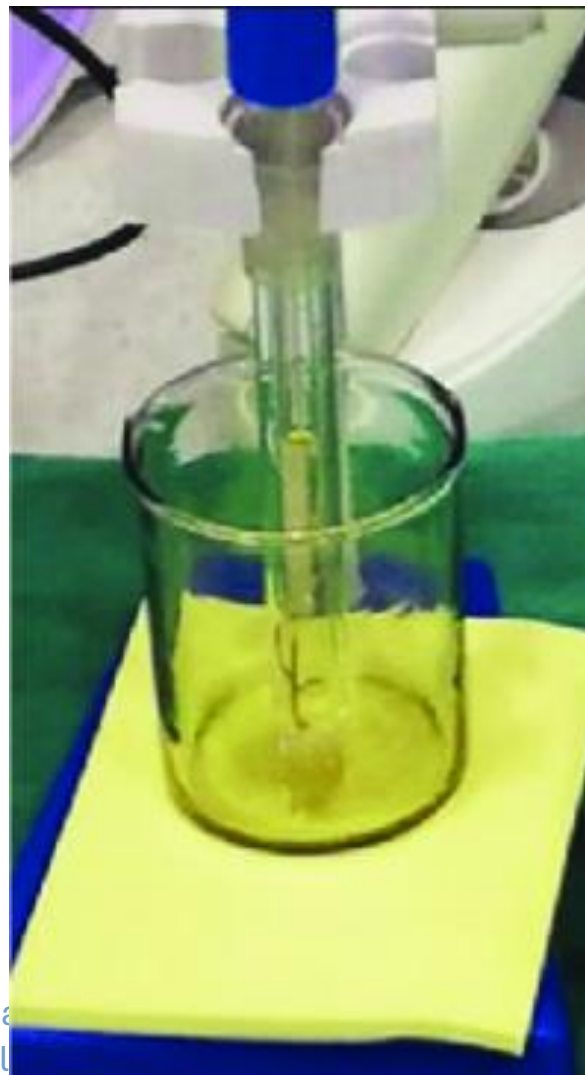
- Is the pH the same everywhere in the mouth?
 - Why? Remember the EPS of the biofilm?
- Where do we want to measure?
- How can we manage the pH in that site?

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Measuring pH



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Buffering vs. Neutralizing

- Neutralizing: Adjusting the acidity or alkalinity of an aqueous solution to bring closer to 7.0
 - Putting out a fire after it's already begun
- Buffering: Adding an agent that makes the aqueous solution more resistant to pH changes. Buffers stabilize the pH, making it harder to change even when exogenous acids/bases are added.
 - Making something more fire resistant

Salivary Buffering

- In a healthy patient, teeth are covered by a thin salivary film (0.1-0.7mm).
- This film acts as a fluid layer through which nutrients and waste products are transported and diffused, both over the surfaces of the teeth/mucosa and within the fluid channels of the biofilm. **Flow** is critical to the functions of saliva.
- This includes salivary components that provide a beneficial buffering effect via:
 - Bicarbonate buffering system
 - Phosphate buffering system
 - Protein buffering system

Salivary Buffering - Bicarbonate

- The most beneficial salivary buffering system between pH 5-7
- Specially active during **stimulated** salivary flow.
 - When salivary production is stimulated, bicarbonate concentration can increase up to 60x compared to resting salivary concentrations
 - Acidic foods (non cariogenic gums/lozenges) stimulate salivary production more than sweet foods
 - Bicarbonate buffering is more effective than protein or phosphate buffering. Therefore, stimulated saliva is more protective than resting saliva.
- Plaque processes sucrose into H⁺
 - Bicarbonate neutralizes H⁺ and creates carbonic acid
 - Carbonic acid breaks down into water and carbon dioxide
- Patients with xerostomia, reduced salivary flow, and salivary gland dysfunction experience much higher caries rates than patients with normal salivary flow. Fermentable carbohydrates in these patients result in more frequent and more prolonged drops in pH.



Salivary Buffering - Phosphate

- The most beneficial salivary buffering system between pH 6-8
- Plays a role in resting/unstimulated saliva when flow rates are low
- Less powerful than bicarbonate buffering
- Phosphate is also critical to remineralization reactions



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Salivary Buffering - Protein

- Major buffering system when pH is below 5
 - But relatively low buffering power overall. All salivary buffering below pH 5 is poor
- Salivary proteins can bind to hydrogen ions so they are less reactive
 - Histatins
 - Mucins
 - Proline-rich proteins
 - Albumin
- These salivary proteins can also contribute to
 - Antimicrobial defense
 - Lubrication
 - Pellicle formation
 - Since some salivary proteins were already present in the pellicle, protein buffering (though weak) is more localized to the tooth surface

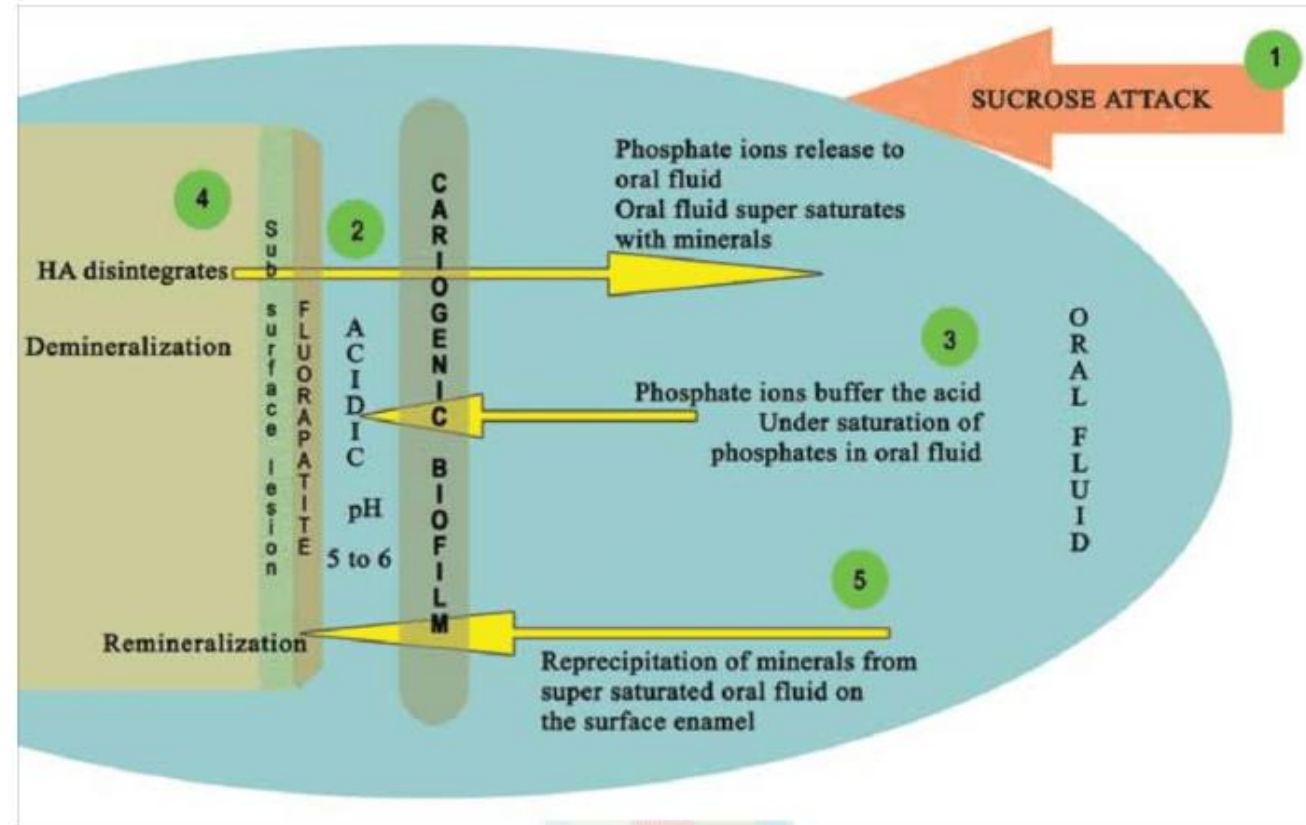


FA Remineralization

We are zoomed in on the enamel surface of a tooth. That enamel is mostly hydroxyapatite. It is coated by a biofilm. The biofilm contains a small amount of bacteria that are capable of fermenting carbohydrates. The biofilm is also well adhered to the tooth surface via an EPS, with fluid channels that facilitate nutrient transport and gradation.

The tooth is also coated with saliva in flow. Saliva contains and transports planktonic bacteria, minerals, buffering agents, and any exogenously introduced substrates such as sucrose.

That saliva has a healthy normal pH of 6.5-7.5, during which time the microbes in the biofilm are in homeostasis (acidogenic bacteria are active but not overgrown).



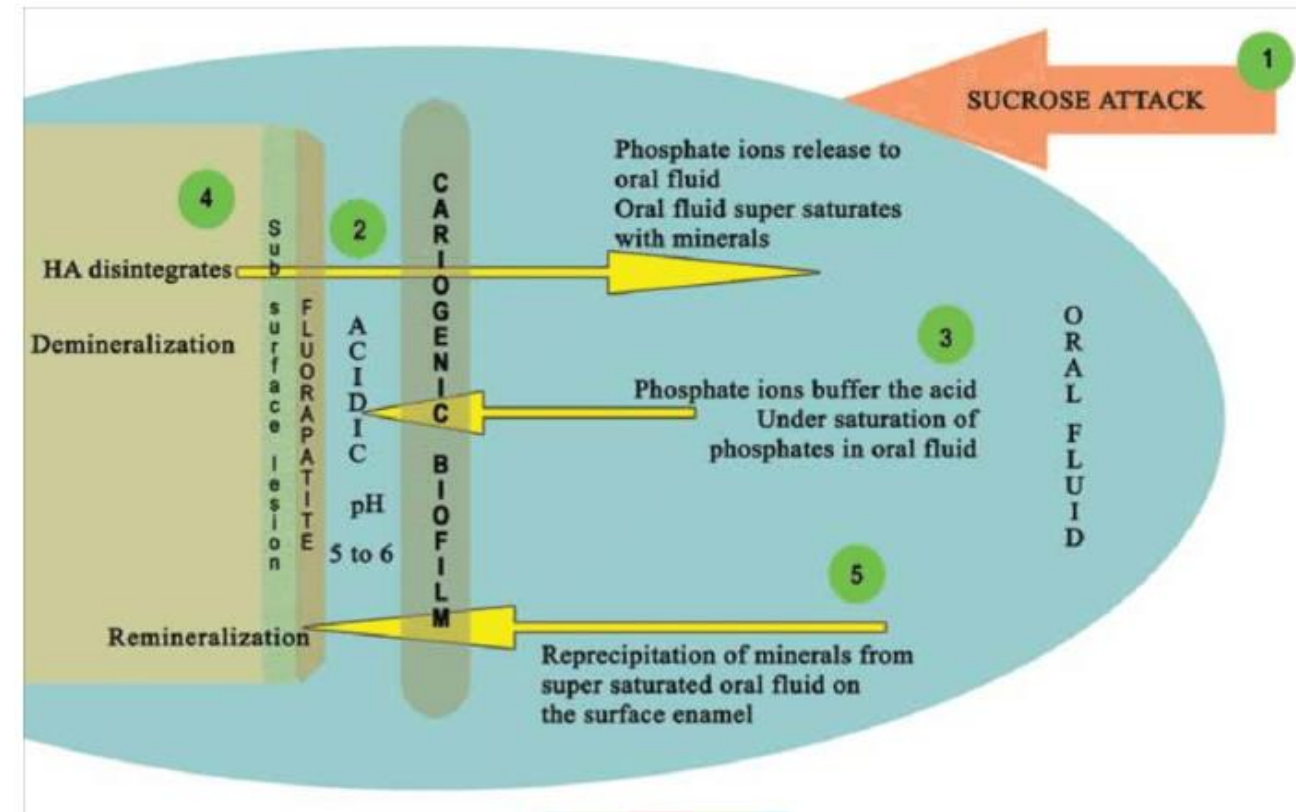
But then:

1. A sucrose attack occurs. [Mission: To advance the science and art of health through education, service, scholarship and social accountability](#)

2. Acidogenic microbes in the biofilm intake sucrose. This increases their growth and metabolic activity. They ferment the sucrose and release acid. This results in microbial succession, selecting for aciduric and acidogenic bacteria, changing the overall distribution of bacteria within the biofilm.

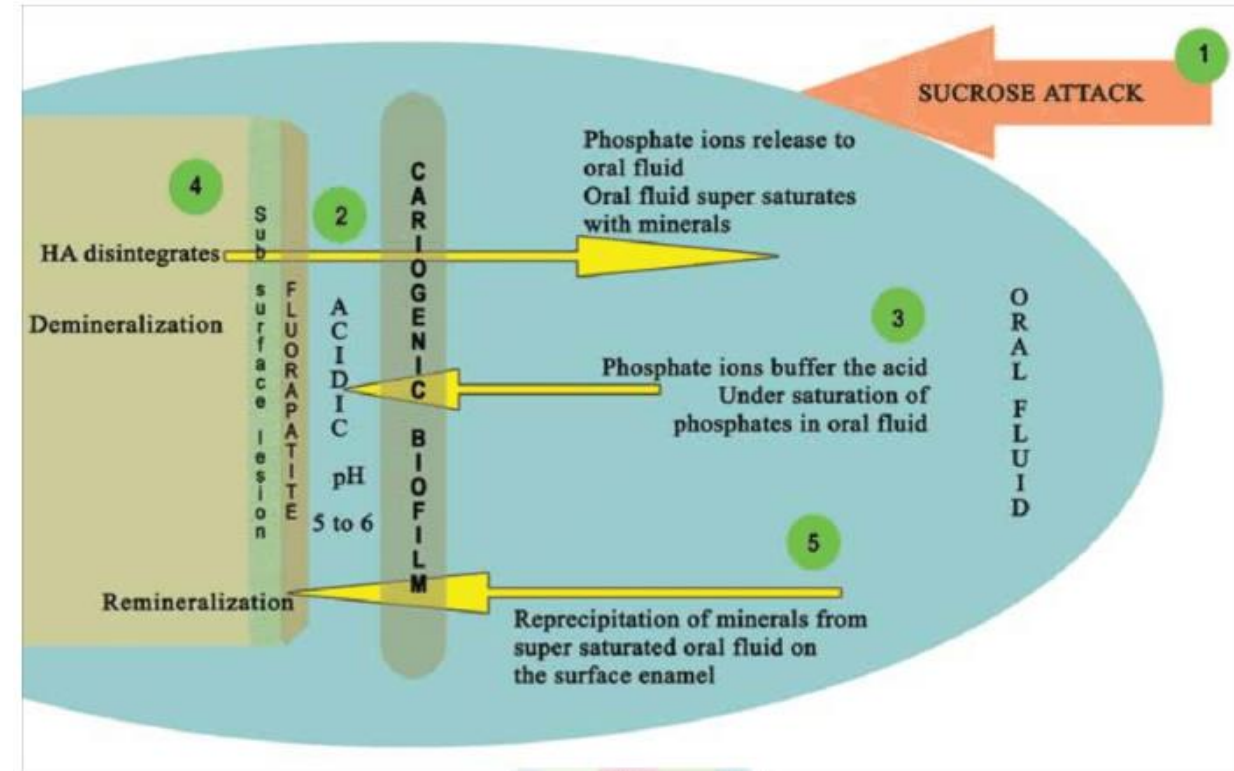
The small zone of salivary fluid between the biofilm and the enamel surface is most affected by the release of this acid, dropping the local pH below 5.5.

3. At physiologic pH in a healthy mouth, the saliva was already supersaturated with minerals such as phosphate. Those phosphates react with the newly produced acidic H^+ ions and provide a buffering affect. However, this **uses up those free phosphates such that the saliva now becomes undersaturated.**



4. **The undersaturation of phosphate in the saliva** and the presence of sufficient H^+ ions results in the demineralization of surface HA into its constituent parts. Buffering may have been insufficient (because the sugar/acid attack was severe) or salivary buffering ions may have become depleted. This releases more phosphate ions (from HA) into the unsaturated saliva, until it becomes supersaturated again.

5. Now that the saliva is supersaturated with apatite building blocks again, those minerals can precipitate onto the surface of the disintegrated enamel to contribute to remineralization (IF the surface pH supports remin). If free fluoride is also present, it will promote the formation of FA over HA. However, FA is only present in the most superficial layers. Remin, whether HA or FA, is limited in its depth of penetration. Therefore, any deeper sub surface lesions will remain.

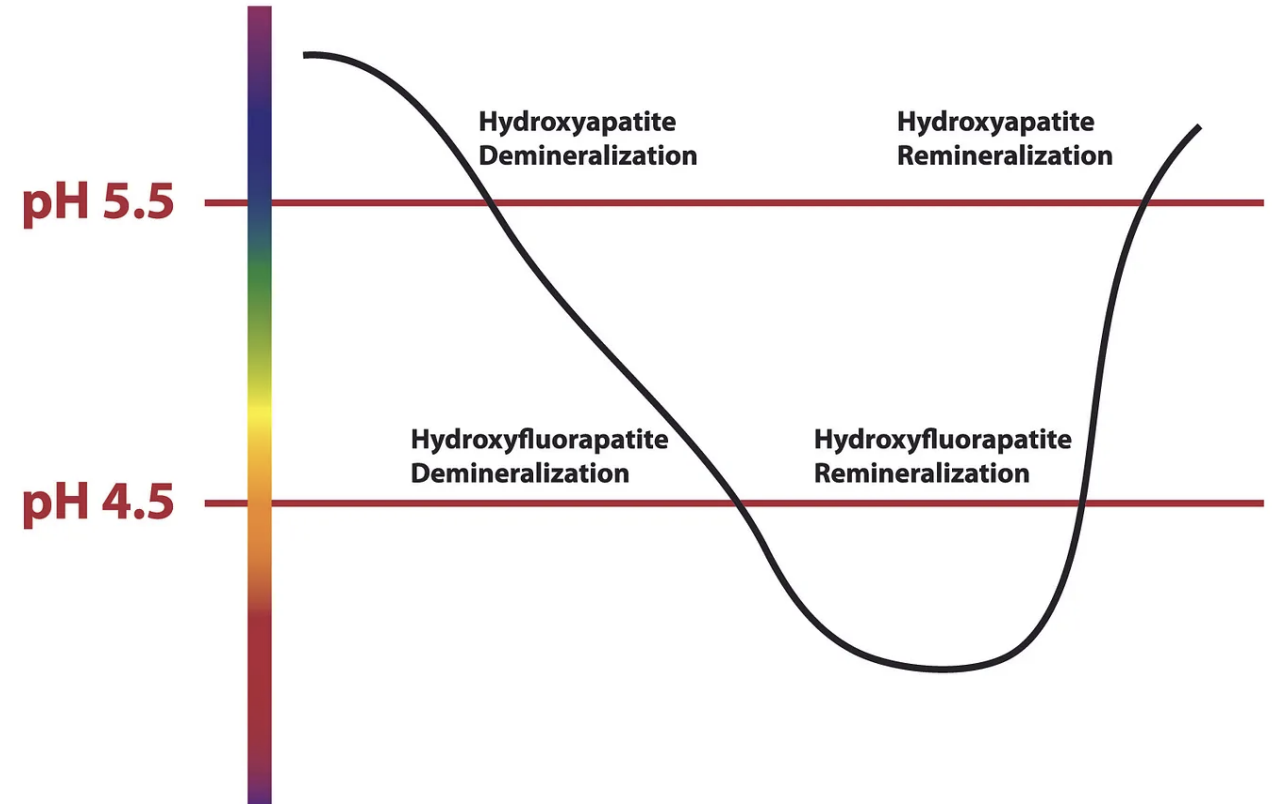


After step 5, the surface of the enamel now contains a higher proportion of FA than it did before.

If the surface of the enamel were 100% FA and 0% HA, that would make the tooth 10x more acid resistant.

So was this demin/remin cycle good or bad?

If we like FA, because it makes the tooth more resistant to sucrose/acid attacks, how do we promote the conversion of more HA to FA? We have to first demin HA, which only occurs in conditions under its critical pH, in order to remin to FA.



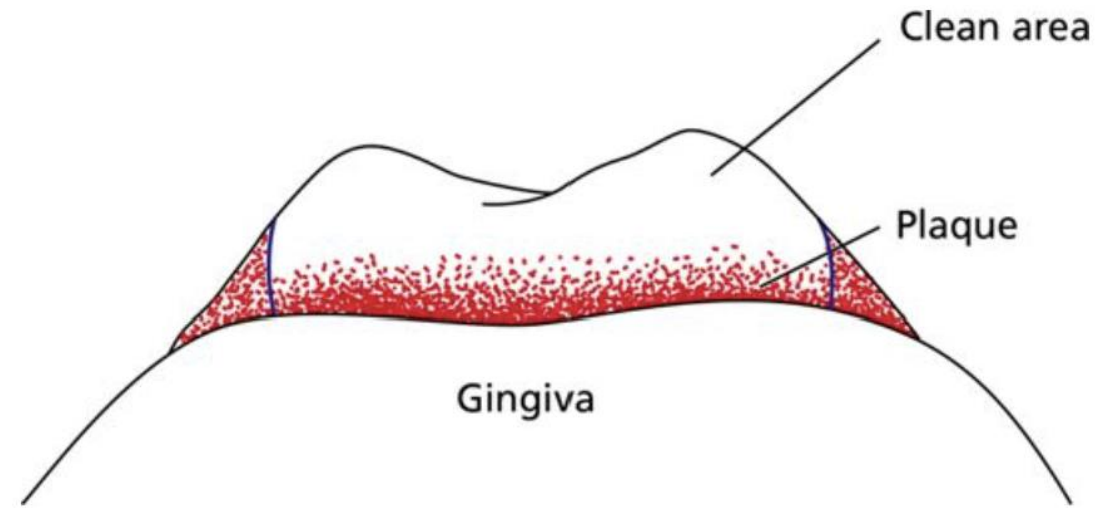


Figure 5.12 Drawing illustrating partly erupted premolar with microbial accumulations predominantly located along the gingival margin.

On eruption, a tooth's enamel is 100% HA.

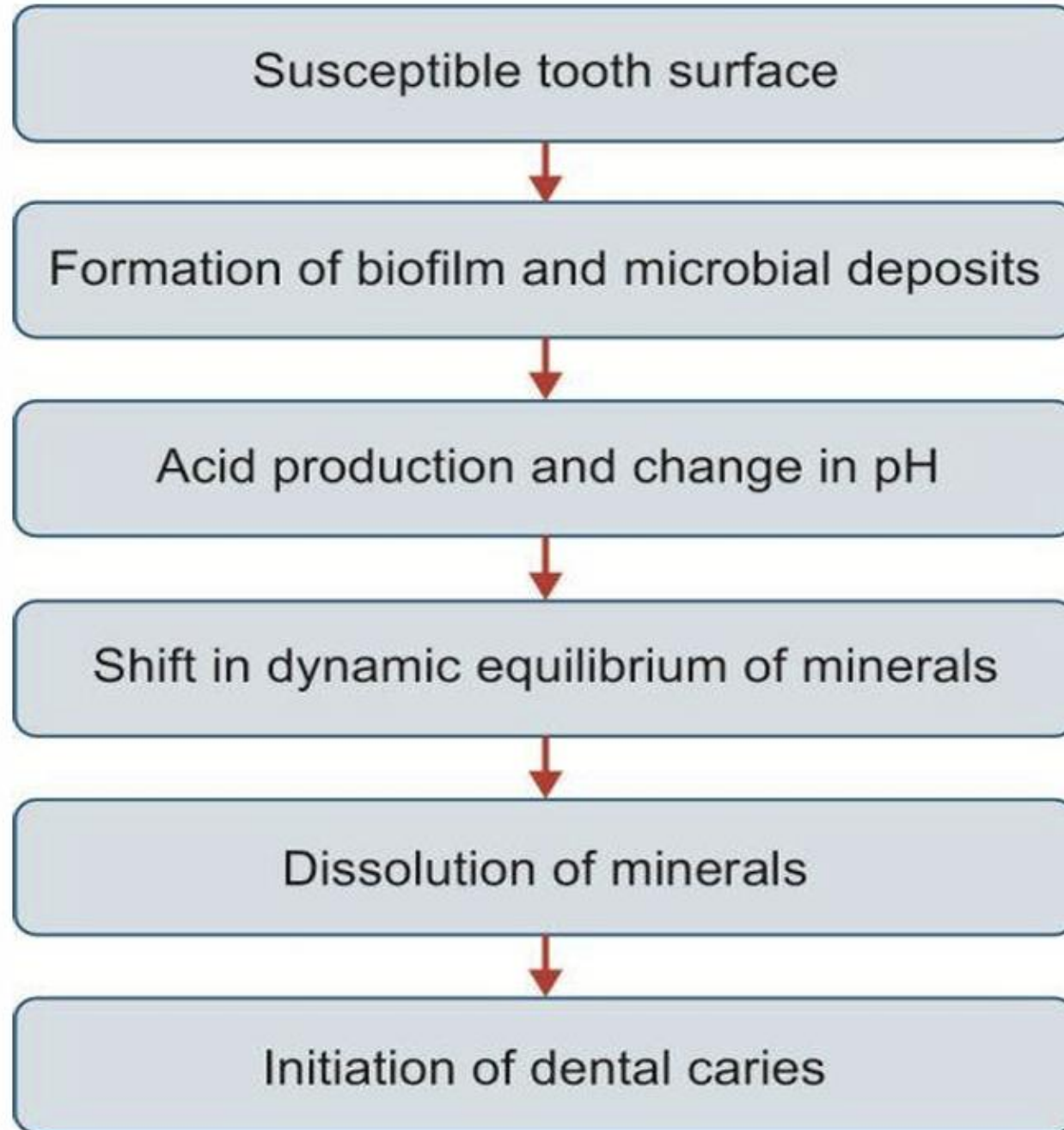
On eruption, a tooth is immediately covered in biofilm.

That biofilm persists undisrupted for often times several months.

- Children do not brush well. Nor do most parents.
- Brushing newly erupted teeth can be uncomfortable.
- Newly erupted teeth are NOT IN FUNCTION. They are not yet in occlusion with opposing dentition.

Until the biofilm on this tooth becomes regularly disrupted (by brushing or by functional occlusion), the demin/remin process will be active. **If the pH stays above 4.5 AND fluoride is present**, this demin/remin process is beneficial as it permits to conversion of HA to FA, **creating a more acid resistant superficial layer of the enamel**. This is part of the post eruptive maturation of teeth.

But if the demin stage persists (without the counterbalance of remin), we see the initiation of dental caries.



What Does Demin Look Like ?

Which parts of this image are DEMIN?

Which parts of this image are REMIN?

Passive remineralization is generally limited to the outer 50-100 microns of enamel (0.05-0.1mm)

Deeper remineralization is rare and requires ideal, prolonged conditions.

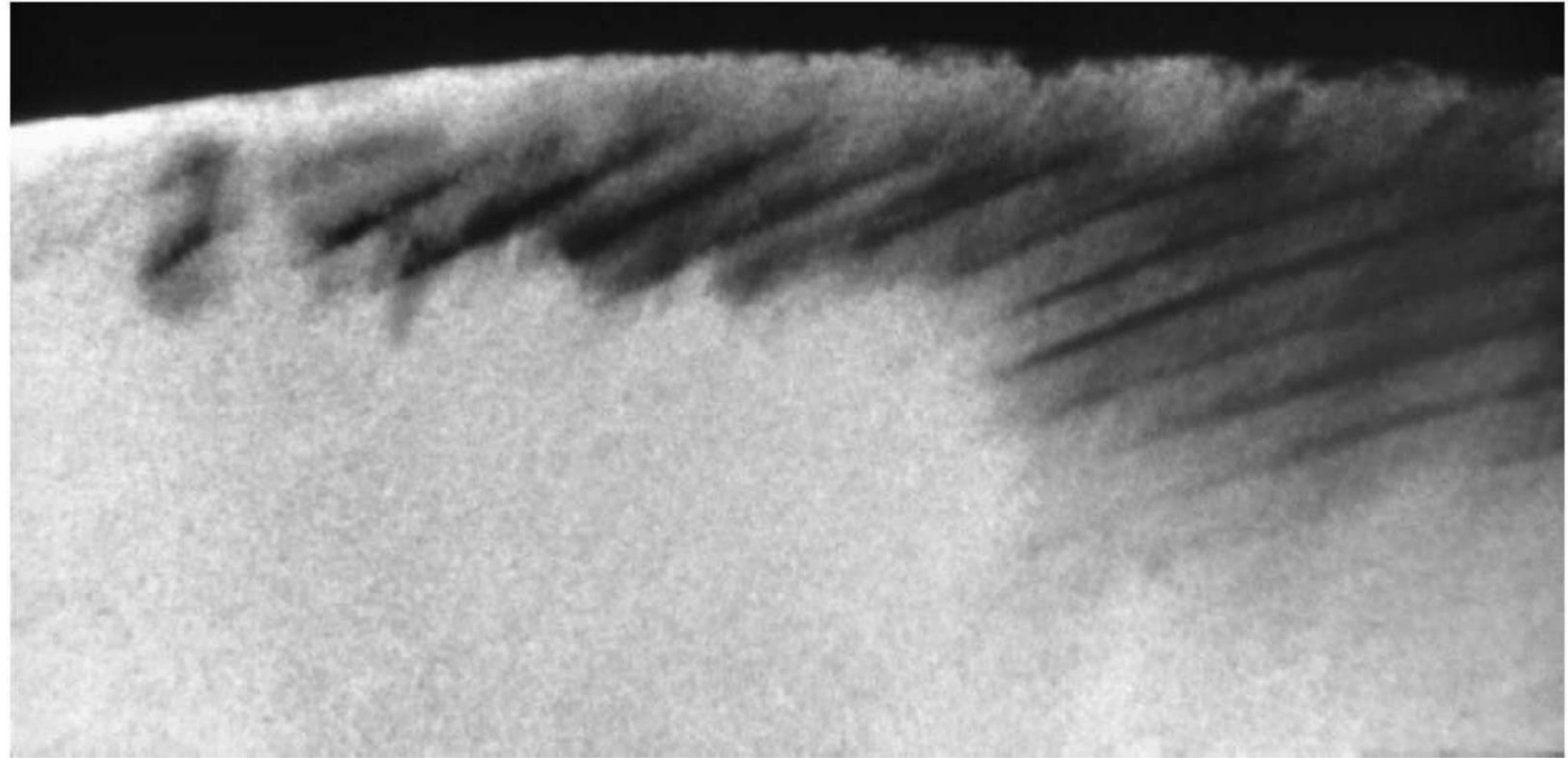


Figure 2.13 When sections are examined by X-rays (microradiography) the calcium loss is directly shown as darker areas in the otherwise white, highly mineralized enamel. Note how the surface enamel varies in mineral content, but is better mineralized than the body of the lesion.

Surface Remineralization

- Fluoride content in the enamel over a subsurface lesion is higher than deeper within the lesion.
- Fluoride content over the subsurface lesion is also higher than the surrounding enamel
- Enamel in newly erupted teeth is 100%HA. Conversion to FA first requires the demin of HA.

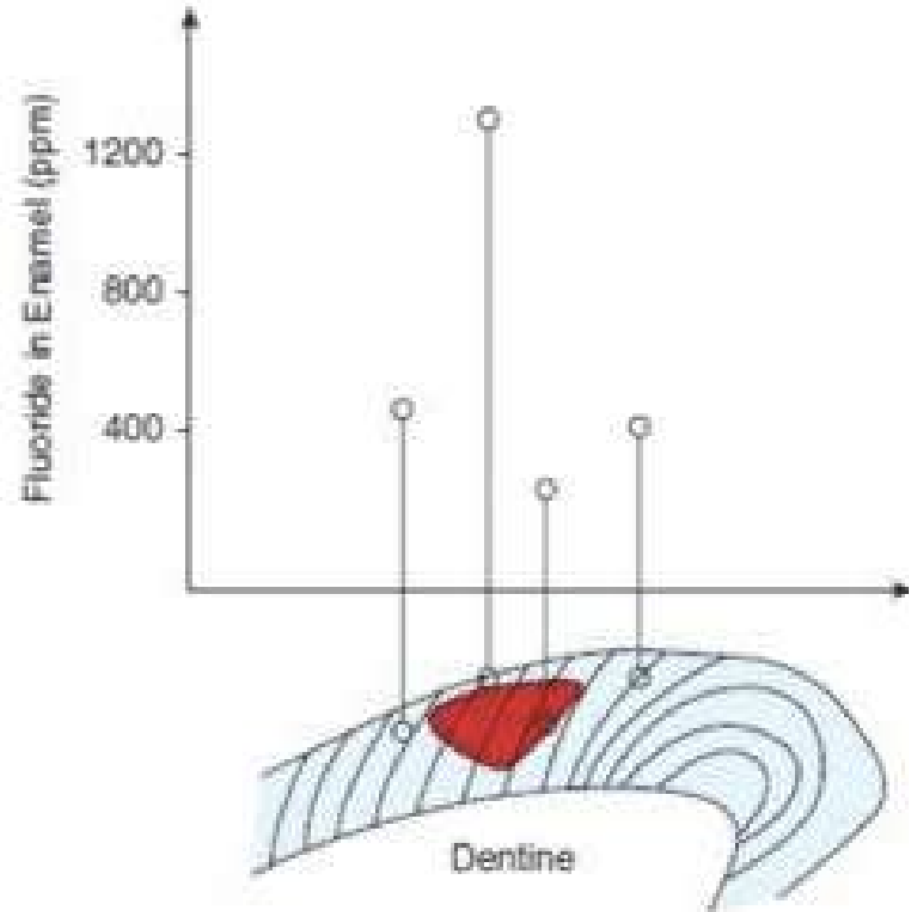


Figure 2.15 Schematic drawing showing the variation in fluoride content in a white spot caries lesion. Note that the highest content is in the surface zone covering the body of the lesion! The ex-

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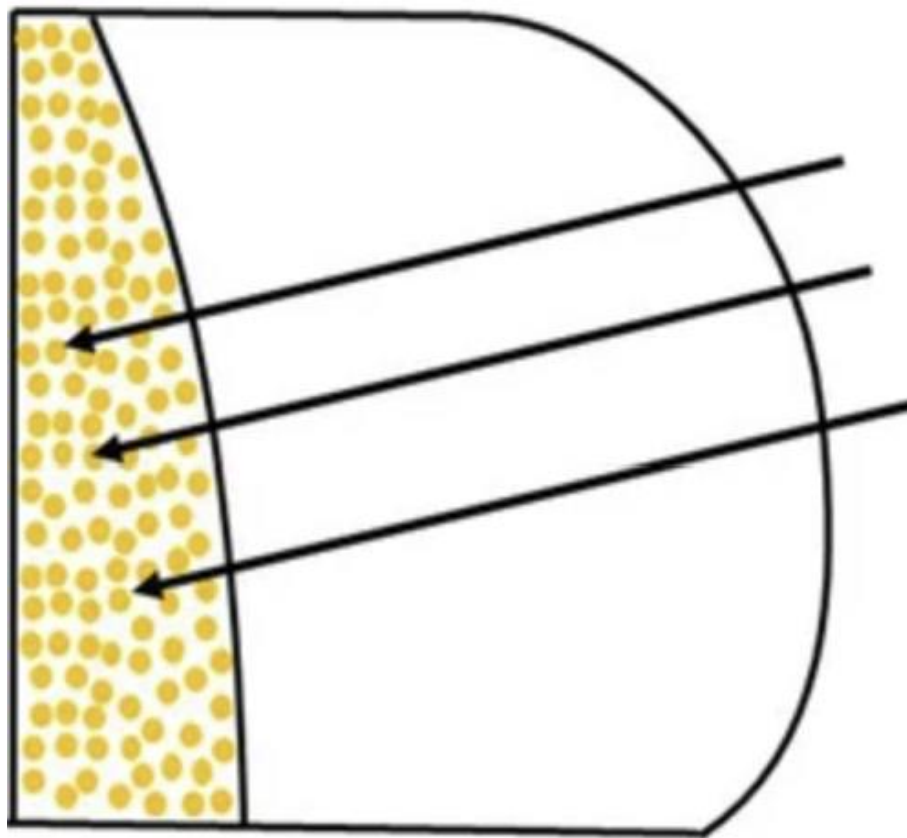
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What Does Demin Look Like Clinically?

- In early stages, a carious lesions
 - Is detectable (by any method) but NOT yet cavitated
 - Can potentially be remineralized to decrease the risk of cavitation
 - But that remineralization has a limit to its depth of penetration
 - Often clinically present as white spot lesions (WSL)
 - Why?
 - Radiographically present as slight radio-opacities

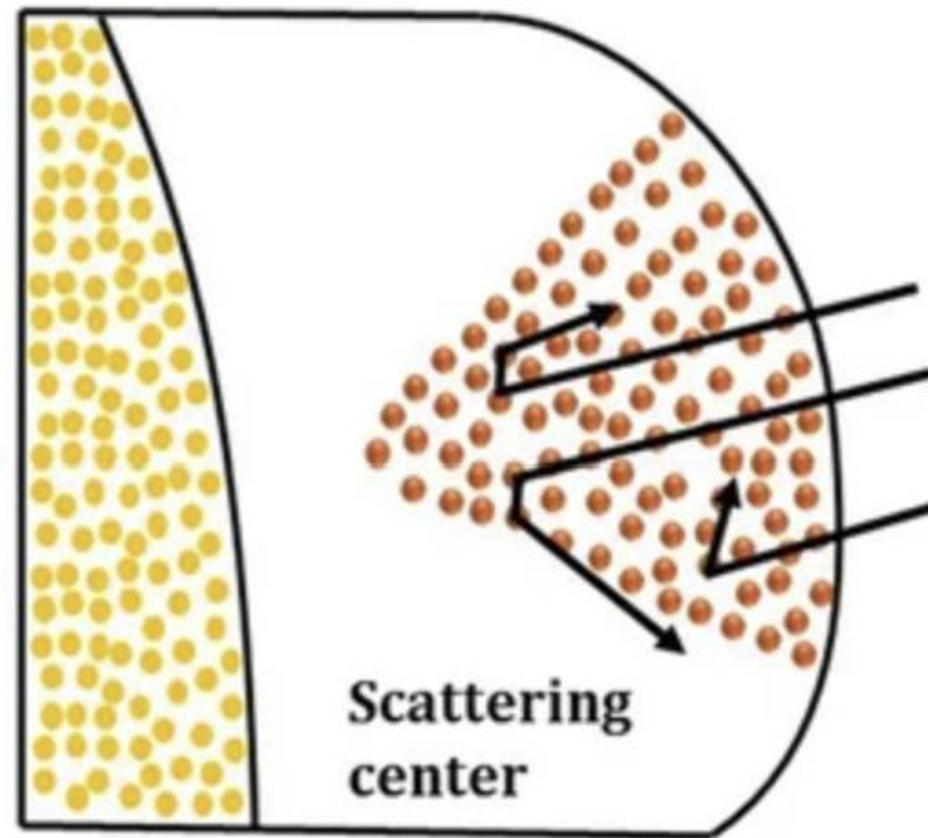


Sound enamel



Dentin Enamel

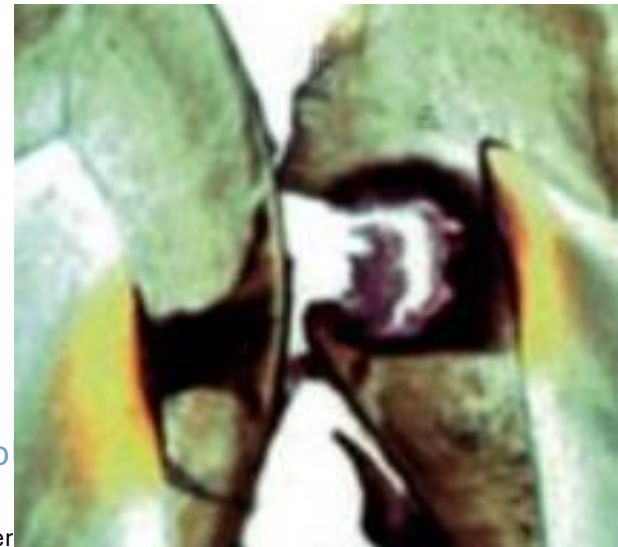
White spot lesions



Dentin Enamel

Demin to Cavitation

- Caries is the result of IMBALANCE between demin and remin (greater demin than remin).
- Enamel is susceptible to cavitation when **30-40%** of its mineral content is lost.
 - o Depends on the forces applied to the demin site
 - o Can take months to years to reach this threshold under normal circumstances
 - o How can you tell what percentage demin a white spot lesion is during an exam?
- Once cavitated, a lesion can not reverse to an uncavitated state
 - o The cavitated lesion CAN remineralize (improve in density of HA/FA and become more acid resistant to further demin)
 - However it will still be cavitated. Remineralization requires an existing framework of structured HA to fill in. It is NOT the de novo formation of new enamel where it does not currently exist.
 - o The cavitated lesion CAN be restored (with a synthetic material)
 - However, restored lesions often develop recurrent caries at the margins if the etiology is not addressed
- Cavitation requires careful visual clinical examination
- Among dentists, take care to use the terms caries vs. cavity correctly!



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Protected Sites

(where "poor oral hygiene" most manifests)

Class 1: pits and fissures (O, B, L)

Class 2 : interproximal smooth surfaces (MOD, etc) of posterior teeth

Class 3: interproximal smooth surface (ML, DF, etc) of anterior teeth

Class 4: involving an incisal corner

Class 5: cervical restoration at the gingival margin

Class 6: involving an incisal edge or cusp tip

A class 1 is called a class 1 because at the time, it was the most prevalent type of restoration needed.

Why is a class 1 more prevalent than a class 2?

Why is a class 2 more prevalent than a class 6?

Why do we see more caries on posterior teeth than anterior teeth?

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Classes	Illustration
Class I	
Class II	
Class III	
Class IV	
Class V	
Class VI	

Protected Site: Pits and Fissures



Name which surfaces of which teeth normally have pits and fissures.

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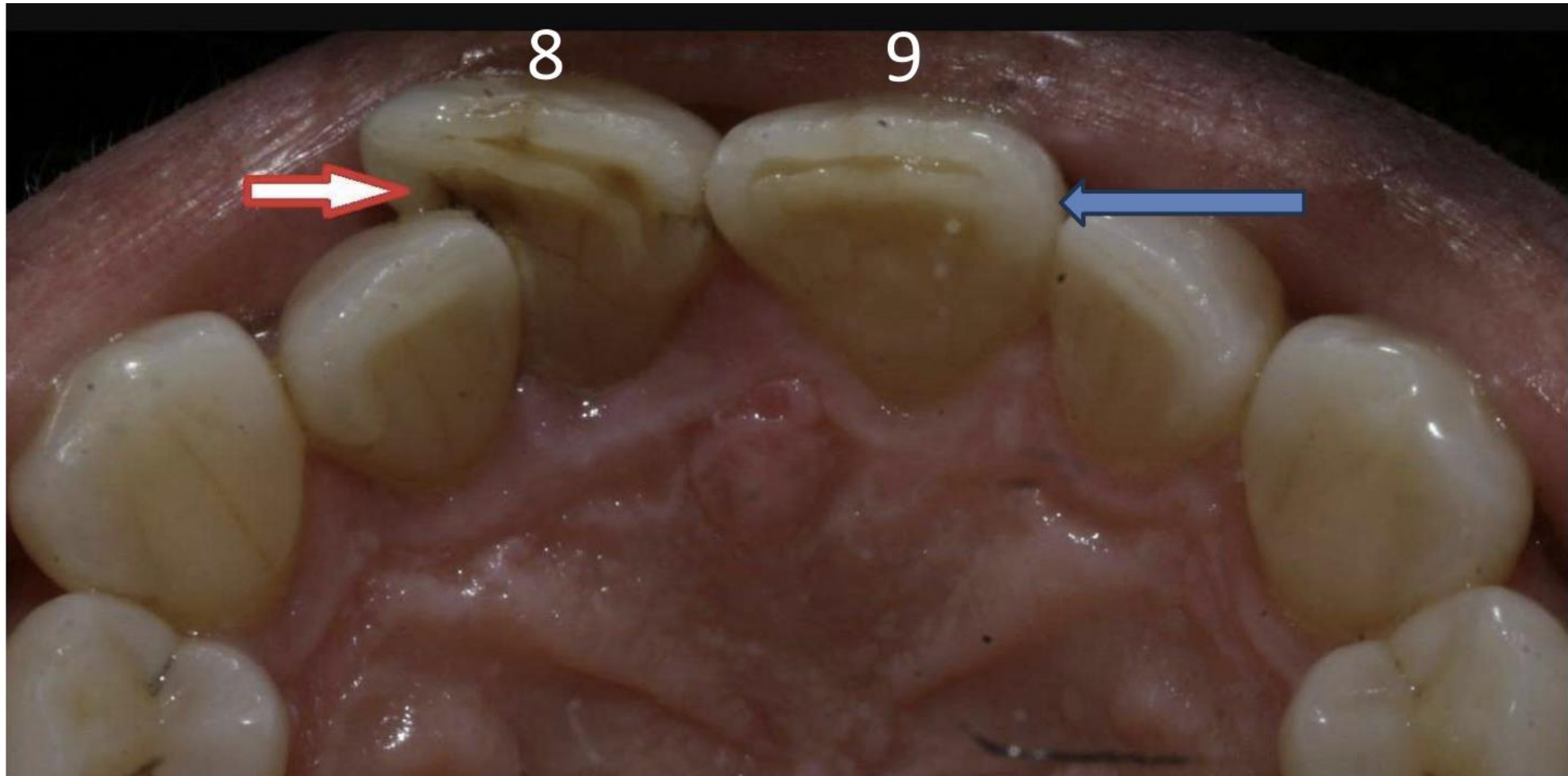
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Protected Site: Overhang

Any poor restorative margin can be a protected site. Whether an interproximal overhang, a shy margin, a rough margin, a poorly bonded margin, a previously sound margin that has since chipped, etc.



Protected Site: Crowding



Protected Sites: Brackets



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Protected Sites: Gingival Margin



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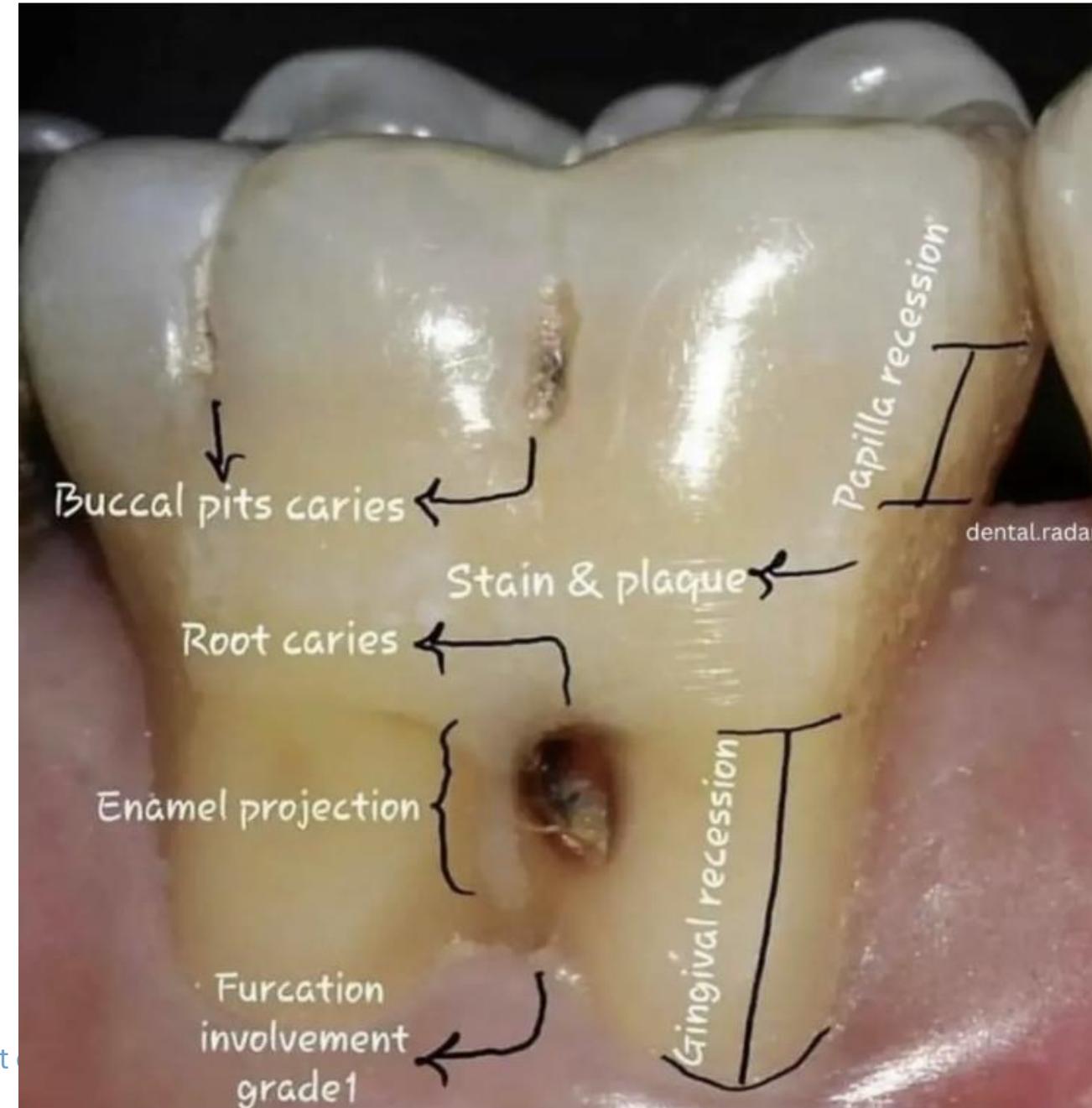
Protected Sites

- Pit and fissures
- Black triangle
- Furcation
- Sensitive/painful to brush
 - o Some roots
 - o Some unerupted teeth



and art

Figure 3. Cleaning furcations.



Protected Sites: Open/Poor Contact



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Protected Sites: Non Functional Surfaces



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Revealing Protected Sites



How do disclosing agents work? What are they made of?

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Plaque Disclosing Agents

- Differentially stain plaque
 - Enamel has low porosity and few organic binding sites, resulting in minimal to no dye absorption
 - Plaque is highly organic and porous, making it an ideal dye binding substrate
 - Some agents use two dyes, allowing patients to visualize mature vs newly formed biofilm
 - "I just ate lunch and didn't brush yet"
 - Erythrosine : stains newer plaque pink/red
 - Brilliant blue: stains old plaque blue/purple
 - Older plaque has
 - Thicker EPS
 - Higher porosity
 - Greater dye retention

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Cones of Decay in Enamel: Smooth Surface

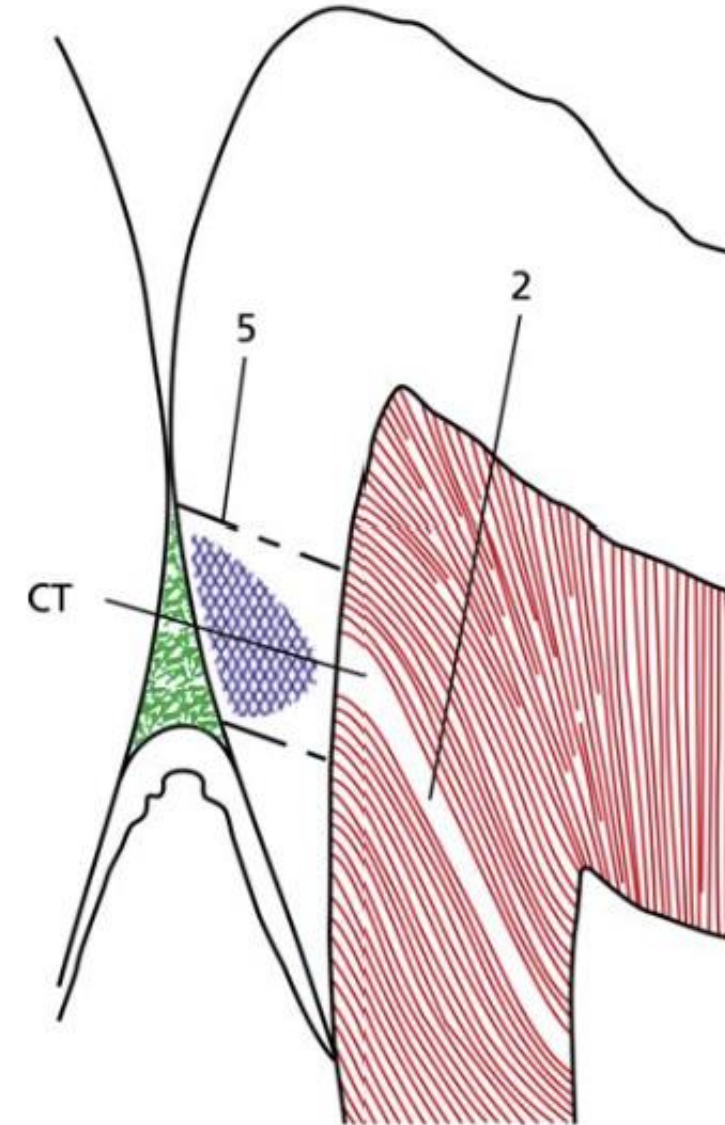
Green = biofilm

Purple = demineralized enamel

This image depicts an interproximal lesion but applies to other smooth surfaces as well (ex: facial and lingual surfaces)

Note the SHAPE of demineralization. Triangular in cross section. Conical in 3D.

For a smooth surface enamel lesion, the widest part of the cone is at the surface. The narrowest part is at the depth of the lesion.



Smooth Surfaces



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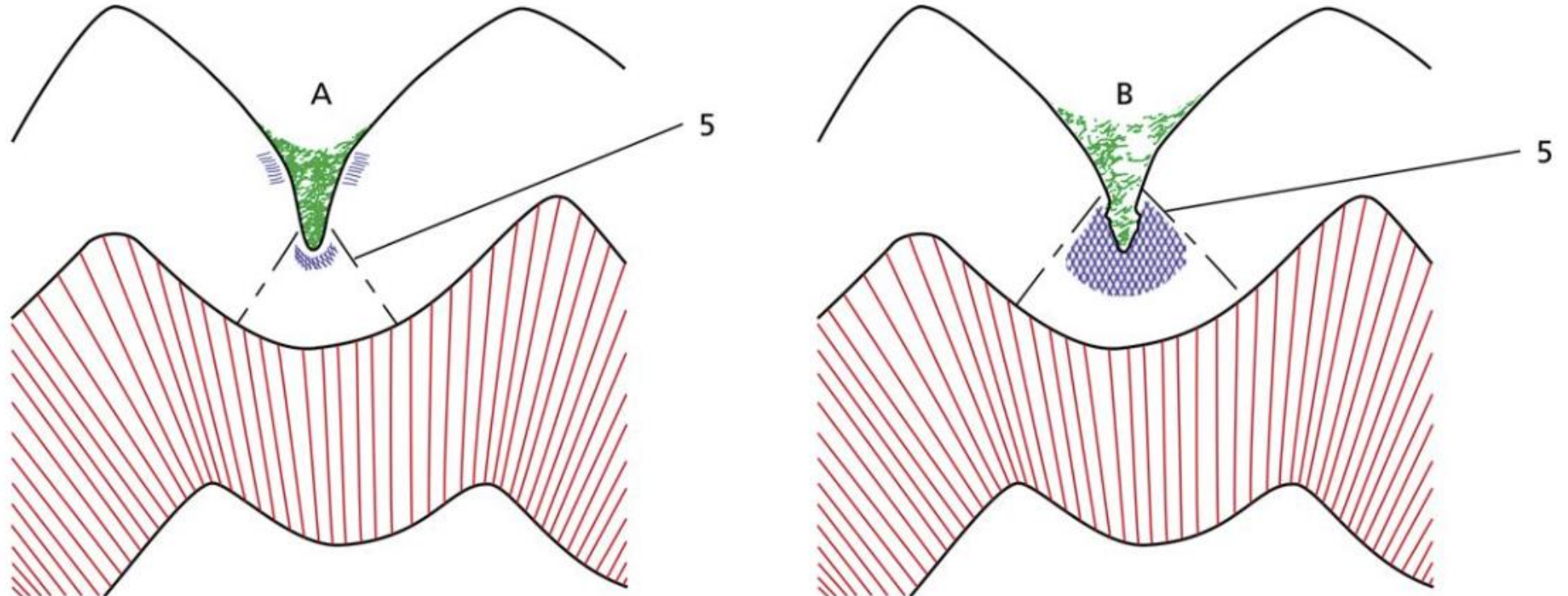
Cones of Decay in Enamel: Pits and Fissures

This image depicts an occlusal groove but applies to other pits and fissures as well (ex: buccal pit, cingulum pit).

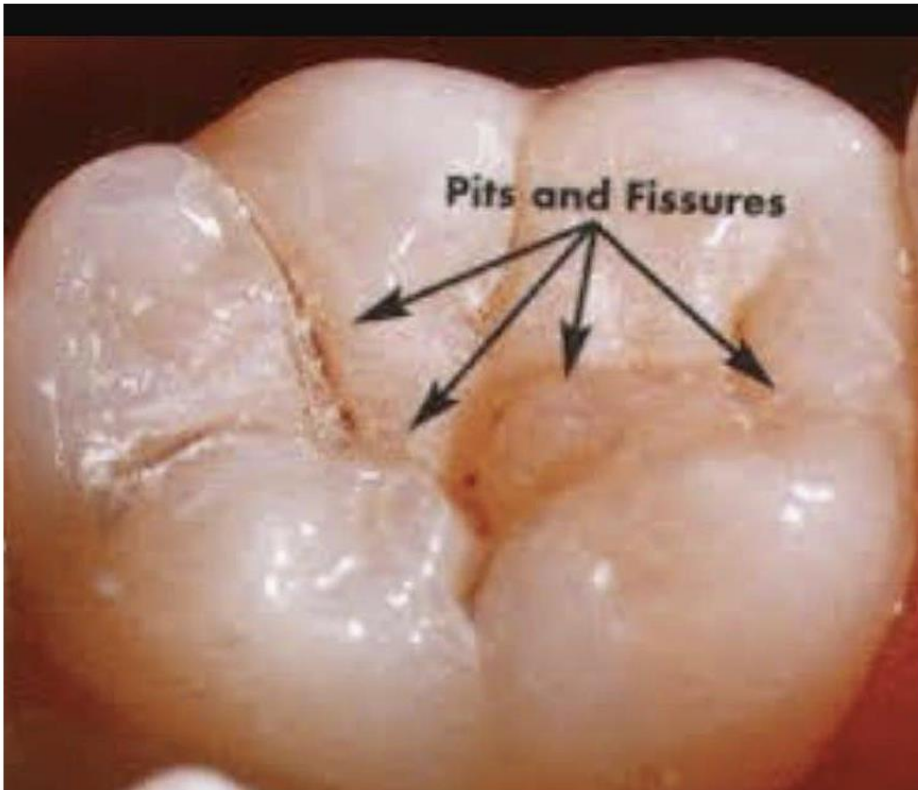
Note the SHAPE of demineralization. For a pit and fissure enamel lesion, the widest part of the cone is at the depth. The narrowest part is at the surface.

The dotted lines (5) depict the direction of the enamel rods in the fissure.

In image B, the depth of the fissure has lost its surface integrity and early cavitation is allowing the ingress of further microbes and their metabolites.



Pits and Fissures

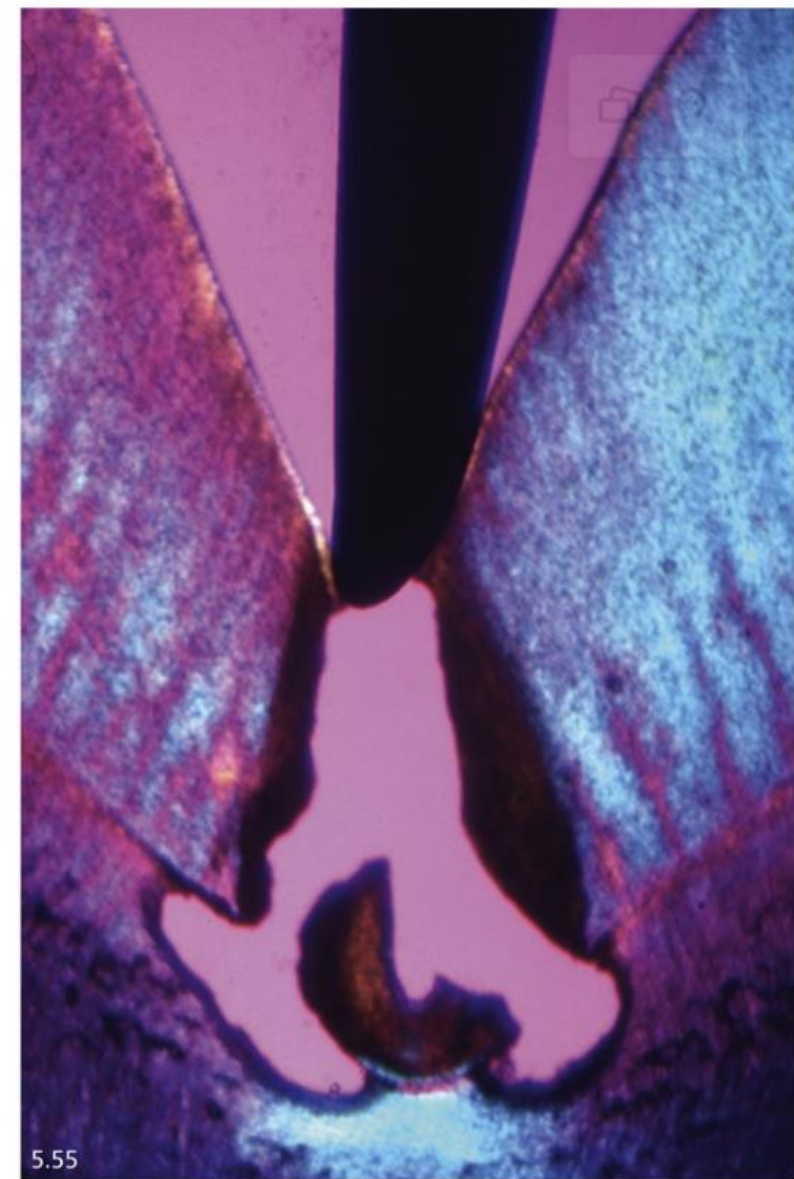


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Conical, though irregular, shape of enamel lesion in a fissure.

Pit and fissure caries are like icebergs, much larger under the surface. Both due to the direction of the enamel rods and due to the potential need to remove any unsupported enamel above the lesion during restoration.



Figures 5.54 and 5.55 Microradiogram and polarized light microscopic illustration of pattern of spread of a caries lesion. At sectioning the tissue was intact! In Fig. 5.55 a probe is located at the entrance to the fissure. Note the apparent undermining character of demineralization. See the text for explanation.

Summary

- Caries **can** develop on ALL surfaces of ALL teeth
- No one surface of normal tooth enamel is chemically or structurally more resistant to developing caries than another
- Caries **does** develop at "protected" sites of the tooth where mechanical disruption is less likely and biofilm accumulation and maturation is more likely to occur
- Dental restorations do not cure caries! Only temporarily TREAT them.

"Caries is a result of the past and present metabolic activities in the microbial deposits covering the tooth surface at any given site. It is a complex disease caused by a complex imbalance in the physiological equilibrium between the dental minerals and the microbiologically induced environmental changes".

Caries is:

- *a result of the past and present (**chronic, cumulative over time**)*
- *metabolic activities (**acidogenic bacteria need to be not only present but actively processing fermentable carbs**)*
- *in the microbial deposits (**biofilm**)*
- *covering the tooth surface (**starts from the outside inward**)*
- *at any given site. (**all enamel is equally susceptible**)*
- *It is a complex disease (**multifactorial etiology, multifactorial points of disruption**)*
- *caused by a complex imbalance (**demin/remin can exist in health without causing caries if it is balanced on both sides**)*
- *in the physiological equilibrium between the dental minerals (**demin/remin is drive by biochemical equilibrium of minerals in the enamel vs. Environment**)*
- *and the microbiologically induced environmental changes (**the acid driving the chemical reaction come from endogenous microbes, not exogenous sources like diet**)*



This assignment does not count toward the final grade.

Lecture 2 Homework

Publish

Assign To

Edit

Submissions accepted as either a .doc, .docx, or .pdf.

Failure to follow instructions or submit on time will result in a 0 for 3-17-26 EPA.

Due Date: APRIL 21st, 2026 9:00am

Note that your dentist may take up to 30 days to share records, hence the extended due date.

Request a copy of your most recent intraoral radiographs from your personal dentist.

At a minimum, we need bitewings of all of your teeth. More images are welcome.

You will be reading your own radiographs near the end of the course. INDIVIDUAL files of each radiograph will be easier to examine and interpret than a composite view with small images.

The date of capture must be noted.

You may remove patient identifiers (name, DOB).

As with all aspects of your individual project, your submission is only visible to yourself and your faculty.

Points 0

Submitting a file upload

File Types doc, docx, and pdf

Due	For	Available from	Until
Apr 21 at 9am	Everyone	-	Apr 21 at 9am

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